

SignalDock

Containerized Security Indicator Intake & Triage Platform on AWS

Console Walkthrough & Configuration Documentation

Region: us-east-1 (N. Virginia)

Prepared by Abdalla Mohamed Aboelfotouh — September 2026

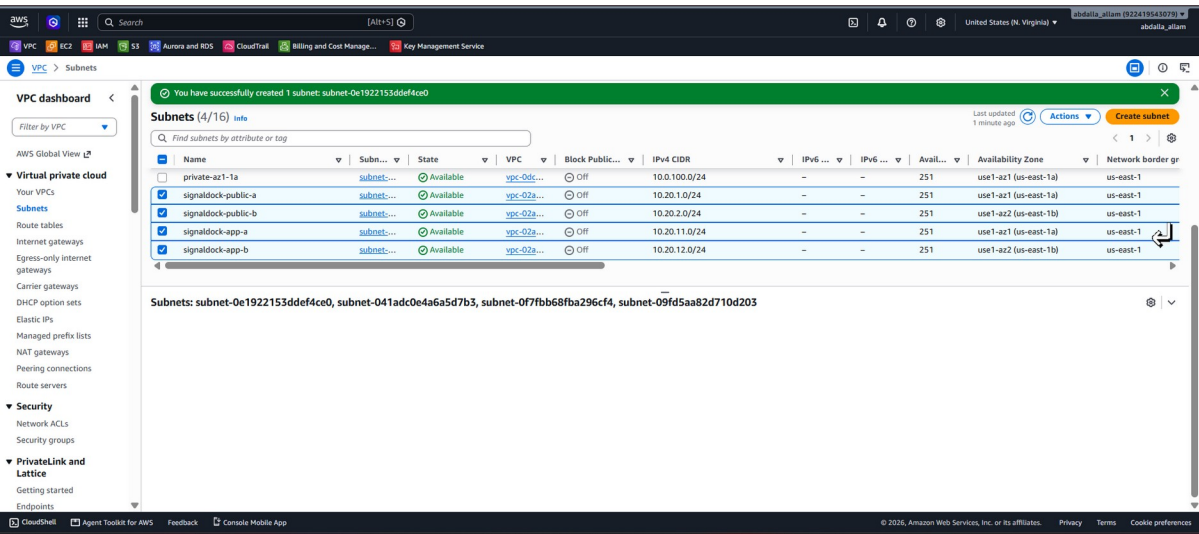
This document walks through the AWS Management Console configuration of SignalDock, a small multi-service application split into a Web, API, and Triage Docker container, each registered in its own ALB target group behind a single path-based Application Load Balancer. Application nodes run in private subnets with no public IP addresses, are managed exclusively through an EC2 Instance Connect Endpoint, persist data in DynamoDB via a Gateway VPC Endpoint, and are kept available through an Auto Scaling Group with both self-healing and CPU-based dynamic scaling, monitored with CloudWatch and alerted through SNS.

Each numbered item below corresponds to one console or terminal screenshot, in the order it was captured, with a short explanation of what it shows and how it fits into the overall design. The screenshots are grouped into fourteen sections that mirror the build workflow: networking, security groups, local Docker development, DynamoDB and its VPC endpoint, the golden-image builder instance, load balancing and target groups, Auto Scaling, end-to-end validation, failure testing, dynamic scaling, alarms with SNS, and a small CI workflow validating the Docker builds.

Section 1 — VPC, Subnets & Routing

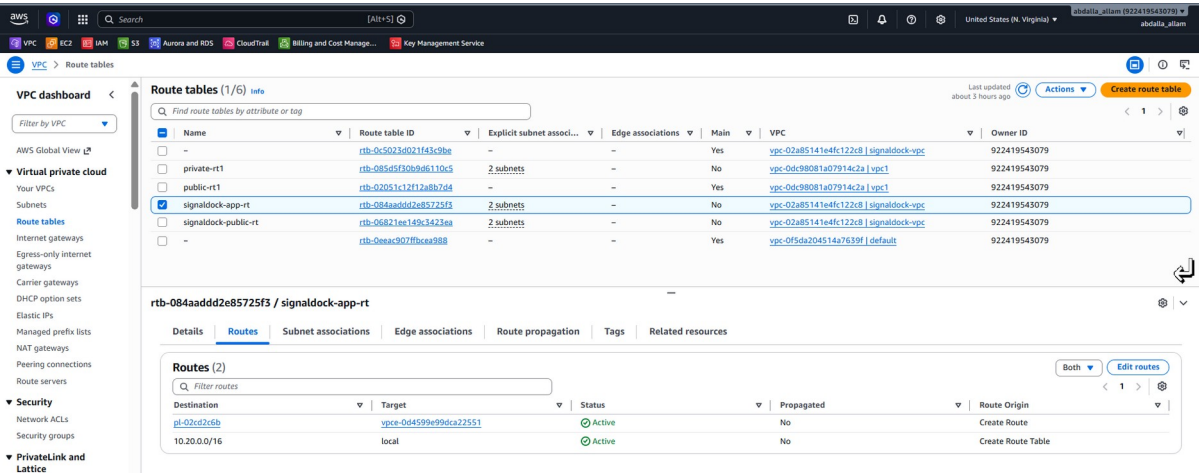
1. Subnets — Created

Confirmation banner after creating the four SignalDock subnets across two Availability Zones.



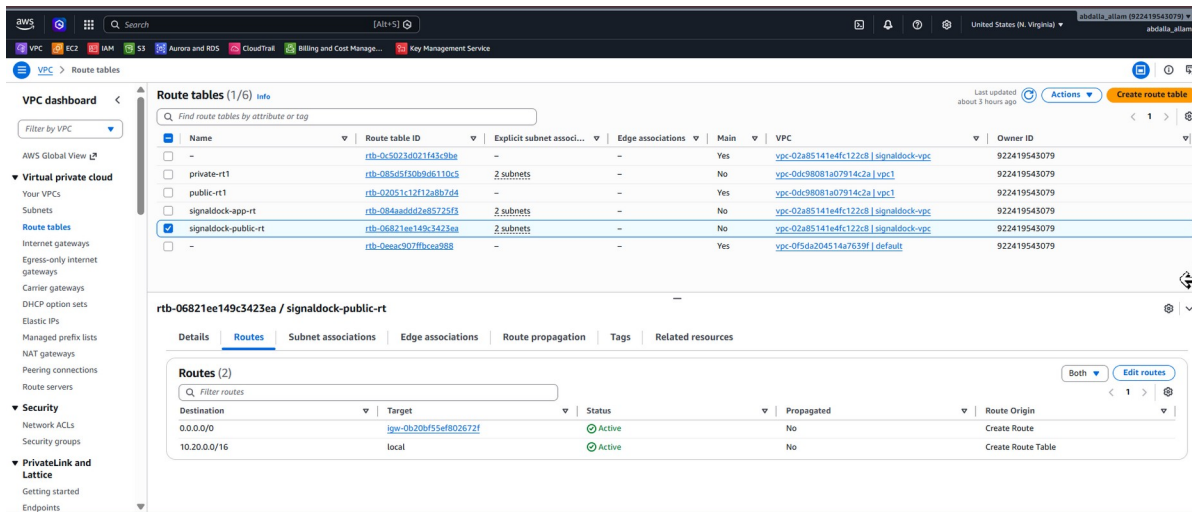
2. Route Tables — Public Selected

Route tables list with signaldock-public-rt selected.



3. Public Route Table — Routes

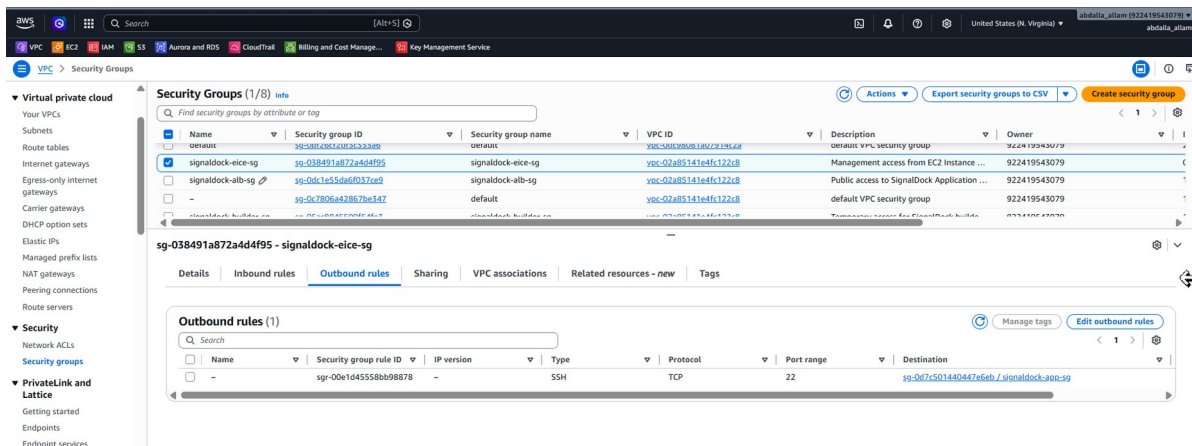
signaldock-public-rt route detail: the local VPC route plus a 0.0.0.0/0 route to the Internet Gateway.



Section 2 — Security Groups

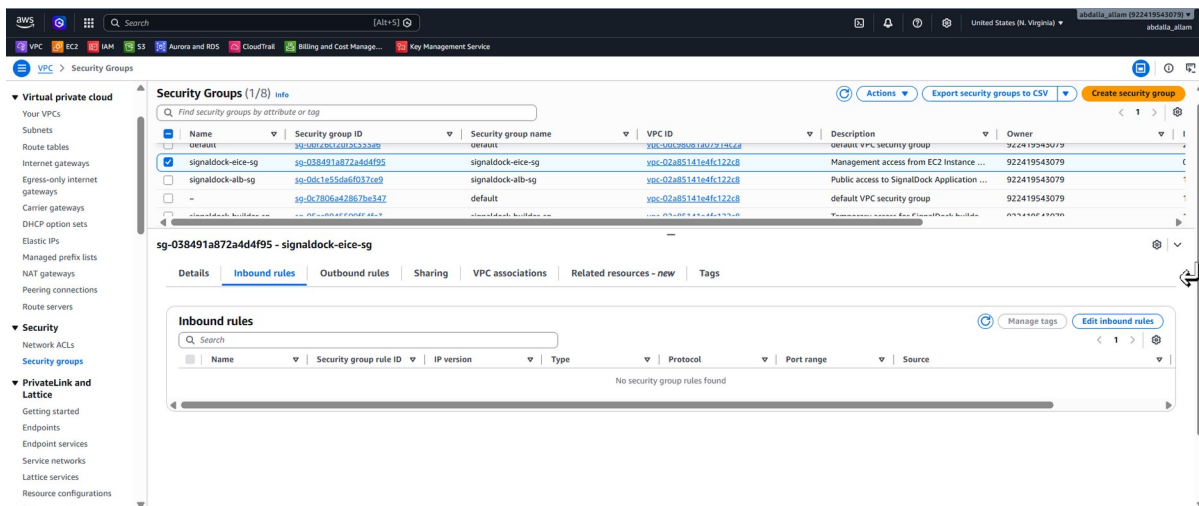
4. EICE Security Group — Outbound

signaldock-eice-sg outbound rules, allowing management traffic to signaldock-app-sg.



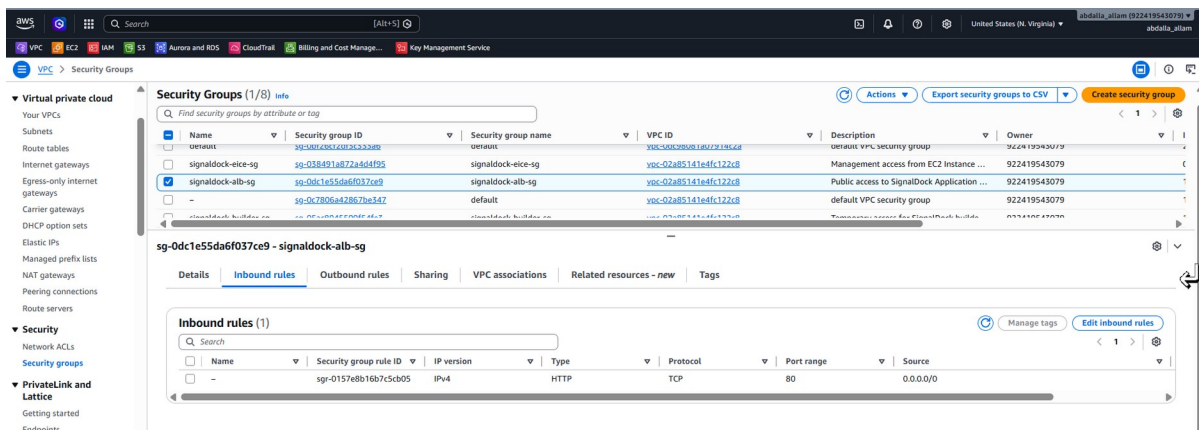
5. Security Groups — ALB SG Selected

Security Groups list with signaldock-alb-sg selected.



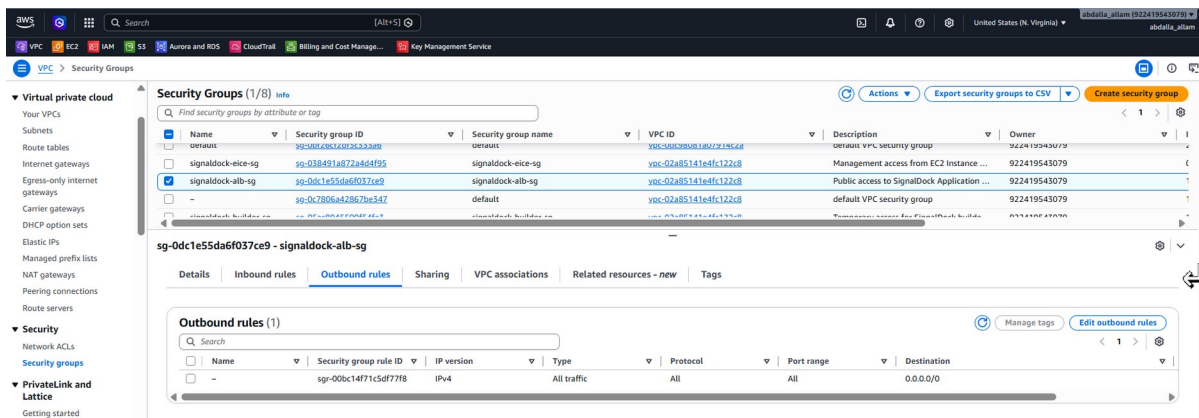
6. ALB Security Group — Inbound

signaldock-alb-sg inbound rules, allowing HTTP on port 80 from the internet.



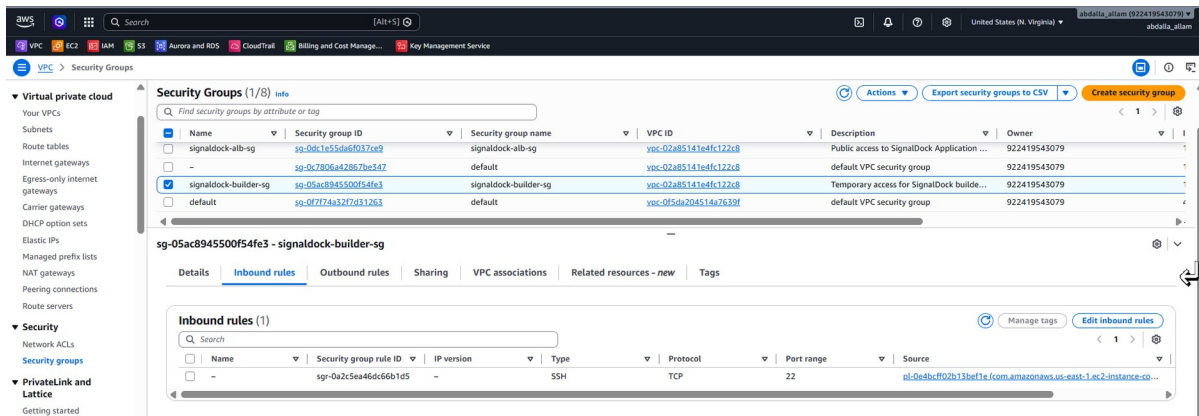
7. ALB Security Group — Outbound

signaldock-alb-sg outbound rules, allowing traffic to signaldock-app-sg.



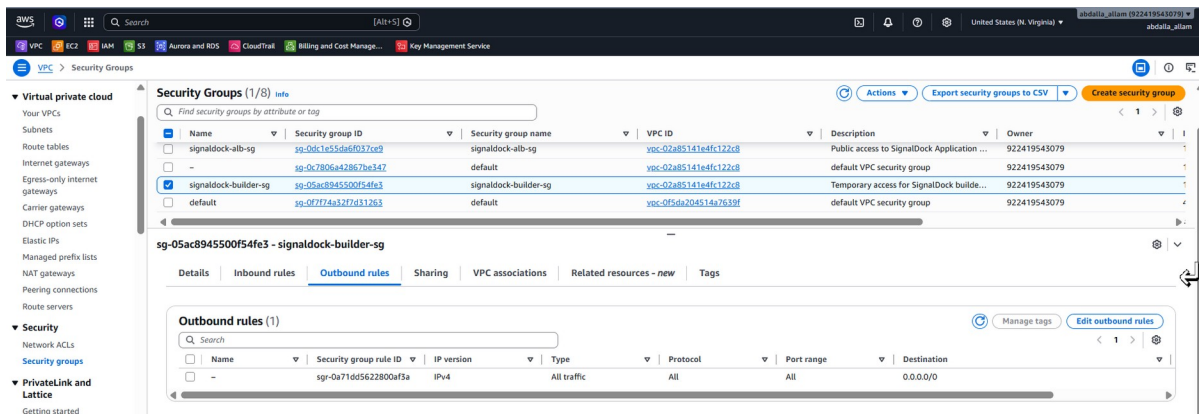
8. Builder Security Group — Inbound

signaldock-builder-sg inbound rules for the temporary builder instance.



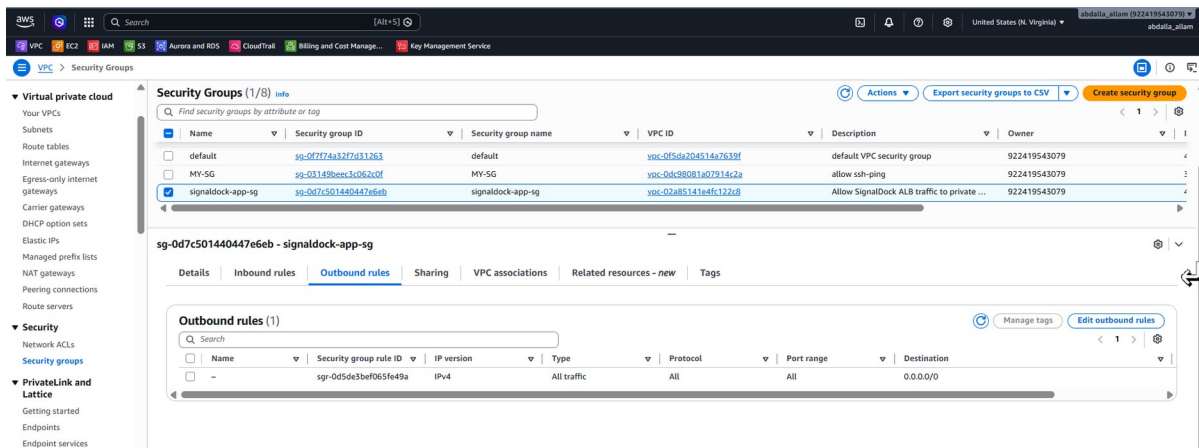
9. Builder Security Group — Outbound

signaldock-builder-sg outbound rules, allowing all outbound traffic.



10. Application Security Group — Outbound

signaldock-app-sg outbound rules.



11. Application Security Group — Inbound

signaldock-app-sg inbound rules: ports 8080/8081/8082 from signaldock-alb-sg, and port 22 from signaldock-eice-sg only.

The screenshot shows the AWS Management Console interface. On the left, the navigation menu includes 'Virtual private cloud' and 'Security'. The main content area is titled 'Security Groups (1/8) info'. A table lists several security groups, with 'signaldock-app-sg' selected. Below the table, the 'Inbound rules (4)' section is expanded, showing a list of rules with columns for Name, Security group rule ID, IP version, Type, Protocol, Port range, and Source. The rules are for ports 8082, 8081, 8080, and 22, all using Custom TCP or SSH protocols.

Name	Security group ID	Security group name	VPC ID	Description	Owner
default	sg-0f7774a32f7d31263	default	vpc-0f5da204514a7639f	default VPC security group	922419543079
MY-SG	sg-03149b6ac3c062c0f	MY-SG	vpc-0dc98081a07914c2a	allow ssh-ping	922419543079
signaldock-app-sg	sg-0d7c501440447e6eb	signaldock-app-sg	vpc-02a85141e4fc122c8	Allow SignalDock ALB traffic to private ...	922419543079

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sg-0cac4aa69d5640eb5	-	Custom TCP	TCP	8082	sg-0dc1e55dae6f037ce9 / signaldock-alb-sg
-	sg-0c8fb723539f7fb66	-	Custom TCP	TCP	8081	sg-0dc1e55dae6f037ce9 / signaldock-alb-sg
-	sg-0ba3cef873c7ed737	-	Custom TCP	TCP	8080	sg-0dc1e55dae6f037ce9 / signaldock-alb-sg
-	sg-041eb540d4c4b268	-	SSH	TCP	22	sg-038491a872a4d4995 / signaldock-slice-sg

Section 3 — Local Docker Build & IAM Policy for DynamoDB

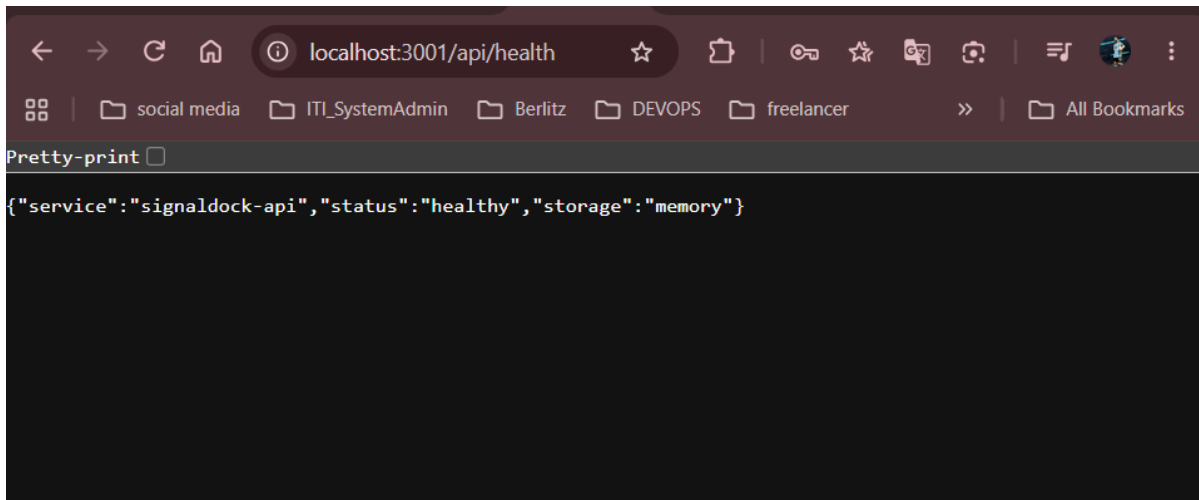
12. Local Docker Build & Compose

Local Windows terminal building and starting the web, api, and triage containers with docker compose.

```
Windows PowerShell
=> CACHED [web 5/5] COPY . . 0.0s
=> [api 3/5] COPY package*.json ./ 0.2s
=> [trriage] exporting to image 0.4s
=> => exporting layers 0.0s
=> => exporting manifest sha256:9495ef7c3efad6f09183c334aa5a5878089a 0.0s
=> => exporting config sha256:a377b210ec617941977d89a6a7792eea25fc68 0.0s
=> => exporting attestation manifest sha256:bcf848daf0e214d7708da836 0.1s
=> => exporting manifest list sha256:ad53ba8680792aba05d4df7b913bce8 0.1s
=> => naming to docker.io/library/signaldock-triage:latest 0.0s
=> => unpacking to docker.io/library/signaldock-triage:latest 0.0s
=> [web] exporting to image 0.4s
=> => exporting layers 0.0s
=> => exporting manifest sha256:50c2ecd007915837d520216e999fea0c5afd 0.0s
=> => exporting config sha256:8068d106833a89c8fdcb39cf1c07d5165d0d1b 0.0s
=> => exporting attestation manifest sha256:ce6be3d3ce85030e50da0ca2 0.2s
=> => exporting manifest list sha256:053ecf43bc012e5d1f7af156deeff95 0.1s
=> => naming to docker.io/library/signaldock-web:latest 0.0s
=> => unpacking to docker.io/library/signaldock-web:latest 0.0s
=> [api 4/5] RUN npm ci --omit=dev 5.0s
=> [web] resolving provenance for metadata file 0.1s
=> [trriage] resolving provenance for metadata file 0.0s
=> [api 5/5] COPY . . 0.1s
=> [api] exporting to image 1.9s
=> => exporting layers 0.8s
=> => exporting manifest sha256:9f983f1f971ce9523e7e102eaec16e31e1e5 0.0s
=> => exporting config sha256:6b1f5960ab28f21a56261b2c0678d6cb37c4d0 0.0s
=> => exporting attestation manifest sha256:0d120d3bee7253ce28290b9b 0.1s
=> => exporting manifest list sha256:ebc9df2801e53b1b1652ac7bbb55e07 0.0s
=> => naming to docker.io/library/signaldock-api:latest 0.0s
=> => unpacking to docker.io/library/signaldock-api:latest 0.9s
=> [api] resolving provenance for metadata file 0.0s
[+] Running 6/6
✔signaldock-api Built 0.0s
✔signaldock-triage Built 0.0s
✔signaldock-web Built 0.0s
✔Container signaldock-web Started 2.7s
✔Container signaldock-triage Start... 2.6s
✔Container signaldock-api Started 2.5s
PS E:\myproject\signaldock>
```

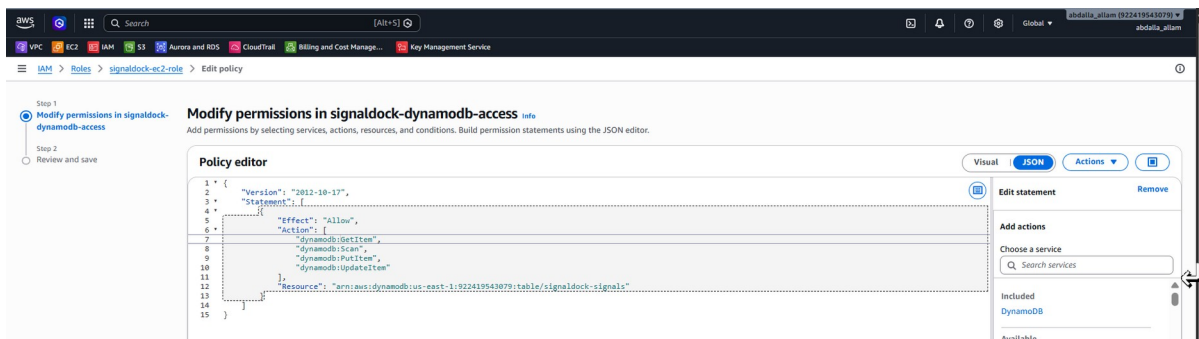
13. Local Health Check — API Service

Browser confirming the API container's /api/health endpoint locally, running in in-memory storage mode.



14. IAM Policy Editor — DynamoDB Access

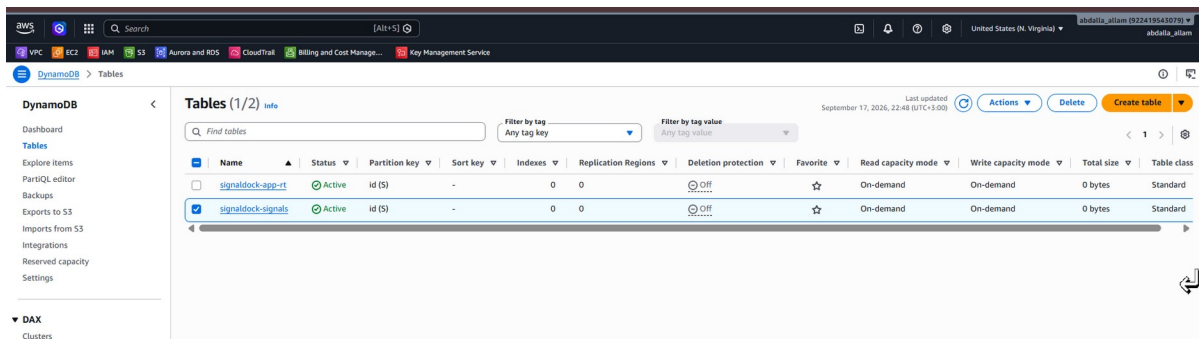
Authoring the signaldock-dynamodb-access IAM policy, scoped to the required table-level actions.



Section 4 — DynamoDB & VPC Gateway Endpoint

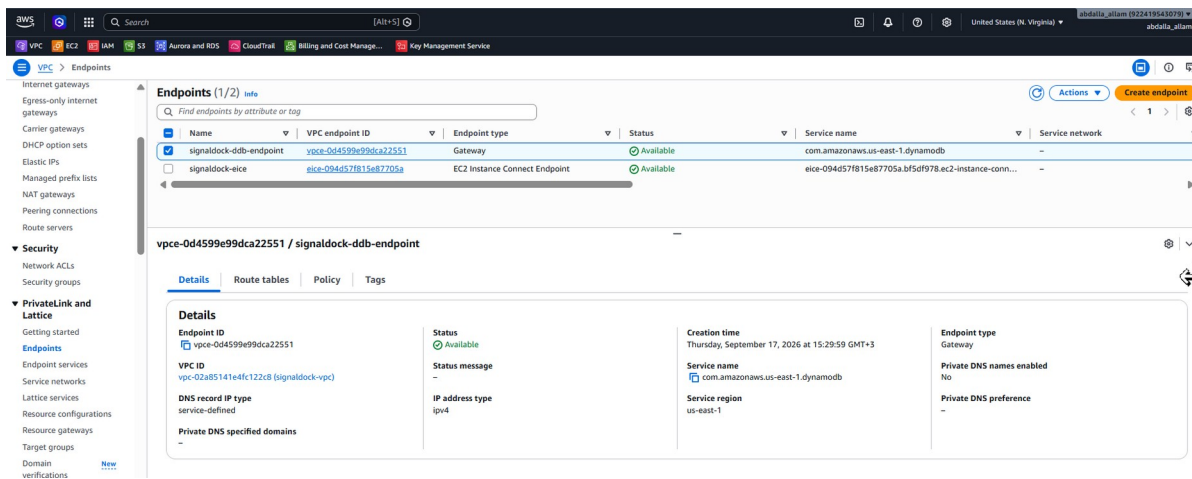
15. DynamoDB — Table Created

DynamoDB console showing the signaldock-signals table.



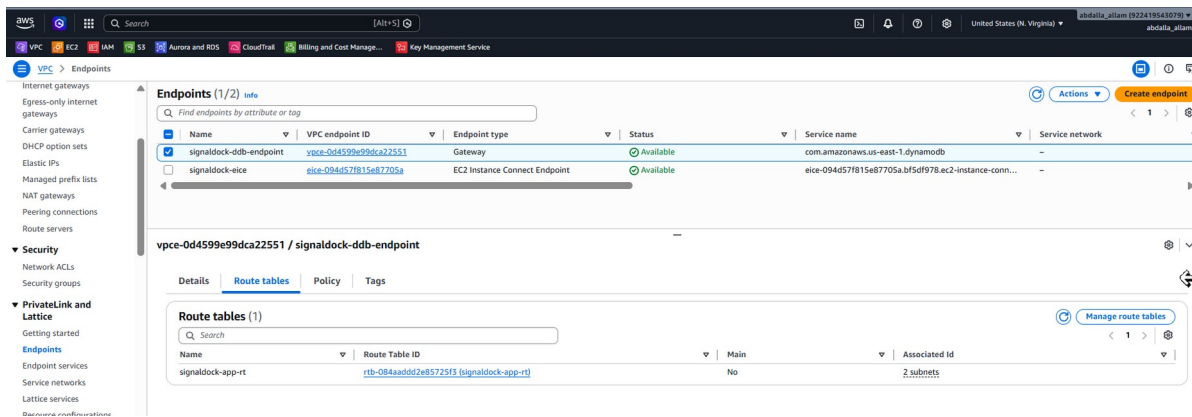
16. VPC Endpoint — DynamoDB Gateway Endpoint

signaldock-ddb-endpoint detail, the Gateway VPC Endpoint providing private access to DynamoDB.



17. VPC Endpoint — Route Tables Tab

DynamoDB Gateway Endpoint's Route Tables tab, confirming its association with the private application route table.



Section 5 — Builder Instance: Container Build & Validation

18. Builder Instance — Container Startup (1)

Builder instance terminal: docker compose build and up output for the three application services.

```
aws | Search [Alt+S] | United States (N. Virginia) | abdalla_allam (922419543079) | abdalla_allam
VPC EC2 IAM S3 Aurora and RDS CloudTrail Billing and Cost Manage... Key Management Service

Running scriptlet: docker-25.0.16-1.amzn2023.0.4.x86_64 19/19
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.

Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch 19/19
Running scriptlet: docker-25.0.16-1.amzn2023.0.4.x86_64 19/19
Verifying : container-selinux-4:2.245.0-1.amzn2023.noarch 1/19
Verifying : containerd-2.2.7-1.amzn2023.0.1.x86_64 2/19
Verifying : docker-25.0.16-1.amzn2023.0.4.x86_64 3/19
Verifying : git-2.50.1-1.amzn2023.0.1.x86_64 4/19
Verifying : git-core-2.50.1-1.amzn2023.0.1.x86_64 5/19
Verifying : git-core-doc-2.50.1-1.amzn2023.0.1.noarch 6/19
Verifying : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64 7/19
Verifying : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64 8/19
Verifying : libcgrouper-3.0-1.amzn2023.0.1.x86_64 9/19
Verifying : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64 10/19
Verifying : libnftnl-1.0.1-19.amzn2023.0.2.x86_64 11/19
Verifying : libnftnl-1.2.2-2.amzn2023.0.2.x86_64 12/19
Verifying : perl-Error-1:0.17030-2.amzn2023.0.1.noarch 13/19
Verifying : perl-File-Find-1.37-477.amzn2023.0.9.noarch 14/19
Verifying : perl-Git-2.50.1-1.amzn2023.0.1.noarch 15/19
Verifying : perl-TermReadKey-2.38-9.amzn2023.0.3.x86_64 16/19
Verifying : perl-lib-0.65-477.amzn2023.0.9.x86_64 17/19
Verifying : pigz-2.5-1.amzn2023.0.4.x86_64 18/19
Verifying : runc-1.3.6-1.amzn2023.0.1.x86_64 19/19

Installed:
container-selinux-4:2.245.0-1.amzn2023.noarch
docker-25.0.16-1.amzn2023.0.4.x86_64
git-core-2.50.1-1.amzn2023.0.1.x86_64
iptables-libs-1.8.8-3.amzn2023.0.2.x86_64
libcgroup-3.0-1.amzn2023.0.1.x86_64
libnftnl-1.0.1-19.amzn2023.0.2.x86_64
perl-Error-1:0.17030-2.amzn2023.0.1.noarch
perl-Git-2.50.1-1.amzn2023.0.1.noarch
perl-lib-0.65-477.amzn2023.0.9.x86_64
runc-1.3.6-1.amzn2023.0.1.x86_64
containerd-2.2.7-1.amzn2023.0.1.x86_64
git-2.50.1-1.amzn2023.0.1.x86_64
git-core-doc-2.50.1-1.amzn2023.0.1.noarch
iptables-nft-1.8.8-3.amzn2023.0.2.x86_64
libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64
libnftnl-1.2.2-2.amzn2023.0.2.x86_64
perl-File-Find-1.37-477.amzn2023.0.9.noarch
perl-TermReadKey-2.38-9.amzn2023.0.3.x86_64
pigz-2.5-1.amzn2023.0.4.x86_64

Complete!
[ec2-user@ip-10-20-1-177 ~]$ git --version
git version 2.50.1
[ec2-user@ip-10-20-1-177 ~]$ docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-10-20-1-177 ~]$
```

19. Builder Instance — Container Startup (2)

Builder instance terminal: container startup logs continuing for all three services.

```
aws [Search] [Alt+S] [United States (N. Virginia)] abdalla_allam (922419543079) abdalla_allam
VPC EC2 IAM S3 Aurora and RDS CloudTrail Billing and Cost Manage... Key Management Service

[ec2-user@ip-10-20-1-177 signaldock]$ docker images
REPOSITORY          TAG         IMAGE ID      CREATED        SIZE
signaldock-triage    1.0         da67cd492ea2  3 seconds ago 172MB
signaldock-api        1.0         884f50d213f3  15 seconds ago 181MB
signaldock-web        1.0         b9fa3846e875  46 seconds ago 172MB
[ec2-user@ip-10-20-1-177 signaldock]$ docker run -d \
--name signaldock-web \
--restart unless-stopped \
-p 8080:8080 \
-e PORT=8080 \
signaldock-web:1.0
ea957dfb4abb1a8c4589a55438afff00d0b989e67c0b0fc78680bf0581bcc237
[ec2-user@ip-10-20-1-177 signaldock]$ docker run -d \
--name signaldock-api \
--restart unless-stopped \
-p 8081:8081 \
-e PORT=8081 \
-e STORAGE_MODE=dynamodb \
-e DYNAMODB_TABLE=signaldock-signals \
-e AWS_REGION=us-east-1 \
signaldock-api:1.0
22f9ffbe0e01621d016a1b9b71d652b7e501260c7705a8b2073914b2c5853589
[ec2-user@ip-10-20-1-177 signaldock]$ docker run -d \
--name signaldock-triage \
--restart unless-stopped \
-p 8082:8082 \
-e PORT=8082 \
signaldock-triage:1.0
2464a5bb4210204513fddb40c2edc720a9ebd6a4ccc80fecacfc77d798eaa100
[ec2-user@ip-10-20-1-177 signaldock]$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
NAMES
2464a5bb4210   signaldock-triage:1.0               "docker-entrypoint.s..." 7 seconds ago  Up 5 seconds  0.0.0.0:8082->
8082/tcp, :::8082->8082/tcp   signaldock-triage
22f9ffbe0e01   signaldock-api:1.0                  "docker-entrypoint.s..." 22 seconds ago  Up 21 seconds  0.0.0.0:8081->
8081/tcp, :::8081->8081/tcp   signaldock-api
ea957dfb4abb   signaldock-web:1.0                  "docker-entrypoint.s..." 30 seconds ago  Up 28 seconds  0.0.0.0:8080->
8080/tcp, :::8080->8080/tcp   signaldock-web
[ec2-user@ip-10-20-1-177 signaldock]$

i-0e64f646e20e4fd47 (signaldock-builder)
PublicIPs: 18.233.200.34 PrivateIPs: 10.20.1.177
```

20. Builder Instance — Health Checks

Builder instance terminal confirming all three containers respond healthy on their respective ports, alongside the instance summary for signaldock-builder.

```
[ec2-user@ip-10-20-1-177 signaldock]$ curl http://localhost:8080/health
ok[ec2-user@ip-10-20-1-177 signaldock]$
[ec2-user@ip-10-20-1-177 signaldock]$ curl http://localhost:8081/api/health
{"service": "signaldock-api", "status": "healthy", "storage": "dynamodb"}[ec2-user@ip-10-20-1-177 signaldock]$
[ec2-user@ip-10-20-1-177 signaldock]$ curl http://localhost:8082/triage/health
ok[ec2-user@ip-10-20-1-177 signaldock]$

i-0e64f646e20e4fd47 (signaldock-builder)
PublicIPs: 18.233.200.34 PrivateIPs: 10.20.1.177
```

21. Builder Instance — Reboot

EC2 Instance Connect session rebooting the builder instance after configuration, to confirm containers restart cleanly.

```
aws [Search] [Alt+S] [Icons] [Settings] United States (N. Virginia) abdalla_allam (922419543079) abdalla_allam
VPC EC2 IAM S3 Aurora and RDS CloudTrail Billing and Cost Manage... Key Management Service
[ec2-user@ip-10-20-1-177 ~]$ curl -X POST http://localhost:8081/api/signals -H "Content-Type: application/json" -d '{"value":"aws-test.example","type":"domain","severity":"high"}'
{"id":"sig-1789658352714","value":"aws-test.example","type":"domain","severity":"high","status":"new","createdAt":"2026-09-17T15:19:12.714Z"}
[ec2-user@ip-10-20-1-177 ~]$ curl http://localhost:8081/api/signals -X PATCH http://localhost:8081/api/signals/sig-1789658352714 -H "Content-Type: application/json" -d '{"status":"investigating"}'
{"status":"investigating","id":"sig-1789658352714"} [ec2-user@ip-10-20-1-177 ~]$
```

22. Builder Instance — API CRUD Test

Builder instance terminal: creating a test indicator via curl POST and updating its status via curl PATCH against the API service.

```
aws [Search] [Alt+S] [Icons] [Settings] United States (N. Virginia) abdalla_allam (922419543079) abdalla_allam
VPC EC2 IAM S3 Aurora and RDS CloudTrail Billing and Cost Manage... Key Management Service
[ec2-user@ip-10-20-1-177 ~]$ sudo reboot
Broadcast message from root@ip-10-20-1-177.ec2.internal on pts/1 (Thu 2026-09-17 15:24:38 UTC):
The system will reboot now!
[]
```

23. Builder Instance — Final Validation

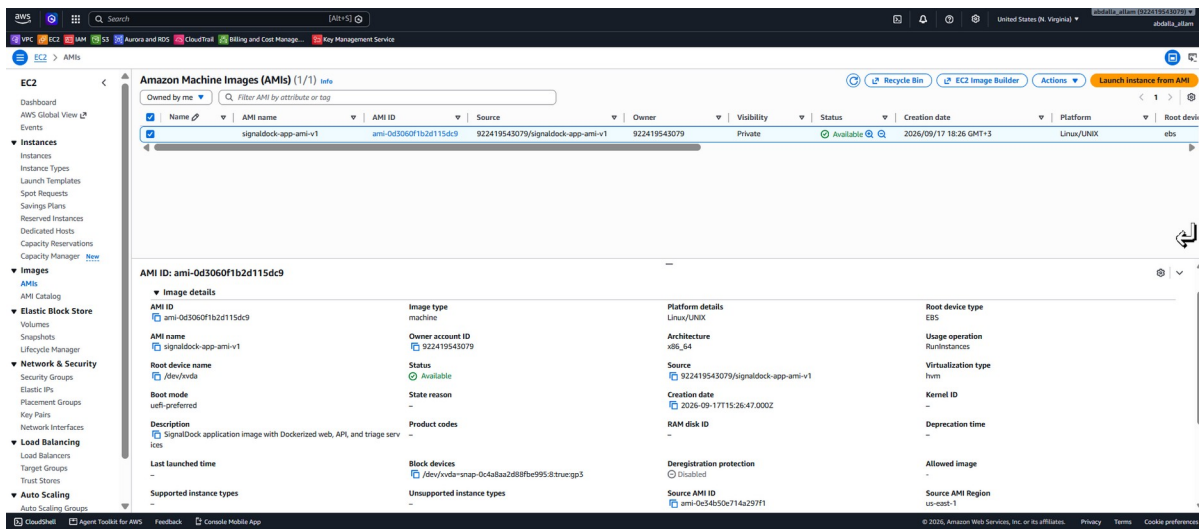
Builder instance terminal: docker ps confirming all three containers are healthy, followed by a final round of endpoint checks before imaging.

```
aws [Search] [Alt+S] [Icons] [Settings] United States (N. Virginia) abdalla_allam (922419543079) abdalla_allam
VPC EC2 IAM S3 Aurora and RDS CloudTrail Billing and Cost Manage... Key Management Service
[ec2-user@ip-10-20-1-177 ~]$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
2464a5bb4210   signaldock-triage:1.0              "docker-entrypoint.s..." 23 minutes ago Up 22 seconds 0.0.0.0:8082->8082/tcp, :::8082->8082/tcp
22f9ffbe0e01   signaldock-api:1.0                 "docker-entrypoint.s..." 24 minutes ago Up 22 seconds 0.0.0.0:8081->8081/tcp, :::8081->8081/tcp
ea957dfb4abb   signaldock-web:1.0                 "docker-entrypoint.s..." 24 minutes ago Up 22 seconds 0.0.0.0:8080->8080/tcp, :::8080->8080/tcp
[ec2-user@ip-10-20-1-177 ~]$ curl http://localhost:8081/api/health
{"service":"signaldock-api","status":"healthy","storage":"dynamodb"} [ec2-user@ip-10-20-1-177 ~]$
[ec2-user@ip-10-20-1-177 ~]$ curl http://localhost:8081/api/signals
[{"createdAt":"2026-09-17T15:19:12.714Z","value":"aws-test.example","id":"sig-1789658352714","severity":"high","status":"new","type":"domain"}, {"id":"sig-179xxxx","status":"investigating"}, {"createdAt":"2026-09-17T15:18:48.377Z","value":"aws-test.example","id":"sig-1789658328377","status":"new","severity":"high","type":"domain"}] [ec2-user@ip-10-20-1-177 ~]$
```

Section 6 — Golden AMI & Launch Template

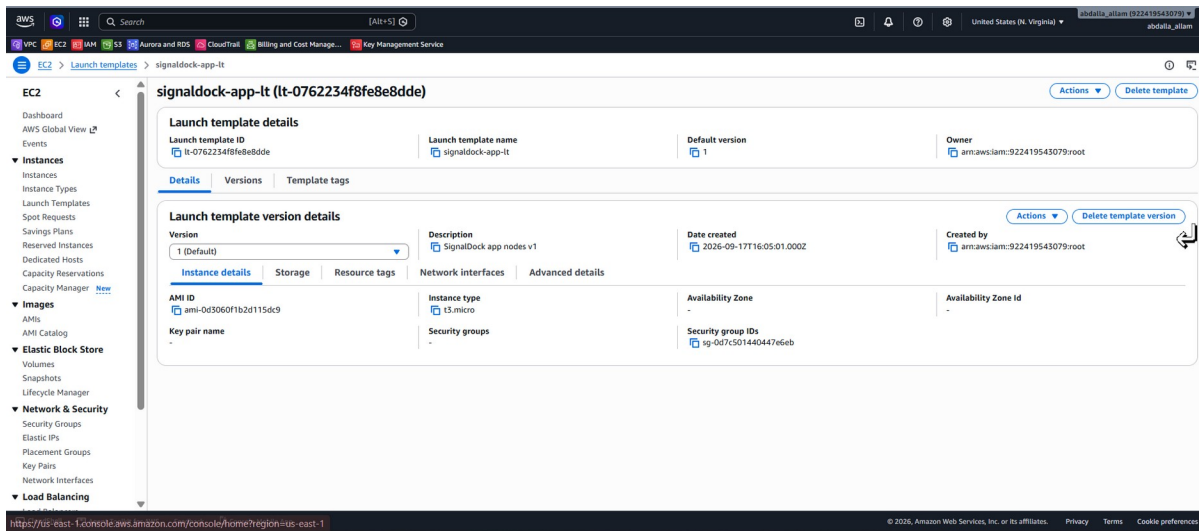
24. Golden AMI Created

AMIs list showing signaldock-app-ami, captured from the fully configured builder instance.



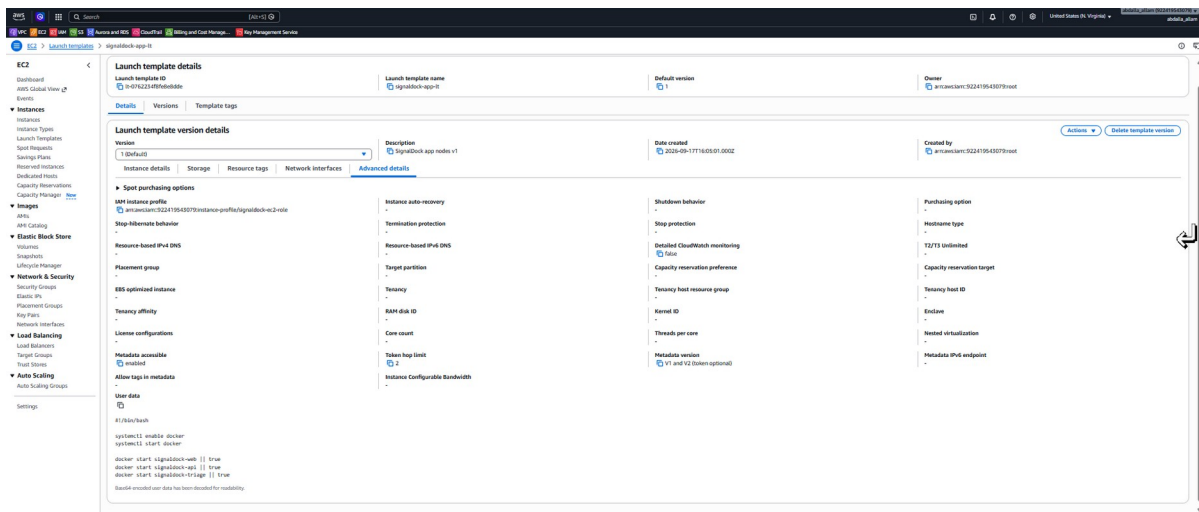
25. Launch Template — Details

signaldock-app-It launch template details.



26. Launch Template — Version Details

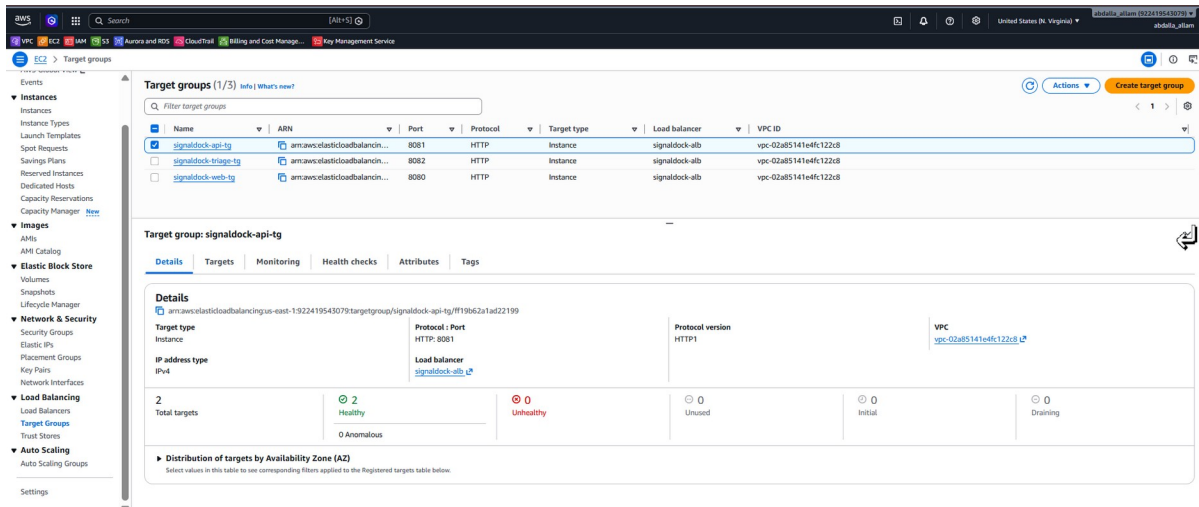
signaldock-app-It launch template version details.



Section 7 — Target Groups & Health Checks

27. Target Groups — API TG Selected

Target groups list with signaldock-api-tg selected.



28. API Target Group — Registered Targets

signaldock-api-tg registered targets, showing both application nodes healthy on port 8081.

The screenshot shows the AWS Management Console interface for Target groups. The left sidebar contains navigation links for various AWS services. The main content area displays a list of target groups under the 'signaldock-api-tg' group. The table below shows the details of the registered targets.

Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
signaldock-api-tg	arn:aws:elasticloadbalancing:us-east-1:922419543079:targetgroup/signaldock-api-tg/1b319433bde91f69	8081	HTTP	Instance	signaldock-alb	vpc-02a85141e4fc122c8
signaldock-triage-tg	arn:aws:elasticloadbalancing:us-east-1:922419543079:targetgroup/signaldock-triage-tg/1b319433bde91f69	8082	HTTP	Instance	signaldock-alb	vpc-02a85141e4fc122c8
signaldock-web-tg	arn:aws:elasticloadbalancing:us-east-1:922419543079:targetgroup/signaldock-web-tg/1b319433bde91f69	8080	HTTP	Instance	signaldock-alb	vpc-02a85141e4fc122c8

Below the table, the 'Health check settings' for the 'signaldock-api-tg' target group are displayed:

- Protocol:** HTTP
- Path:** /api/health
- Port:** Traffic port
- Healthy threshold:** 5 consecutive health check successes
- Unhealthy threshold:** 2 consecutive health check failures
- Interval:** 30 seconds
- Success codes:** 200

29. API Target Group — Health Check Settings

signaldock-api-tg health check configuration, using the /api/health path.

The screenshot shows the AWS Management Console interface for Target groups. The left sidebar contains navigation links for various AWS services. The main content area displays a list of target groups under the 'signaldock-triage-tg' group. The table below shows the details of the registered targets.

Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
signaldock-api-tg	arn:aws:elasticloadbalancing:us-east-1:922419543079:targetgroup/signaldock-api-tg/1b319433bde91f69	8081	HTTP	Instance	signaldock-alb	vpc-02a85141e4fc122c8
signaldock-triage-tg	arn:aws:elasticloadbalancing:us-east-1:922419543079:targetgroup/signaldock-triage-tg/1b319433bde91f69	8082	HTTP	Instance	signaldock-alb	vpc-02a85141e4fc122c8
signaldock-web-tg	arn:aws:elasticloadbalancing:us-east-1:922419543079:targetgroup/signaldock-web-tg/1b319433bde91f69	8080	HTTP	Instance	signaldock-alb	vpc-02a85141e4fc122c8

Below the table, the 'Health check settings' for the 'signaldock-triage-tg' target group are displayed:

- Protocol:** HTTP
- Path:** /api/health
- Port:** Traffic port
- Healthy threshold:** 5 consecutive health check successes
- Unhealthy threshold:** 2 consecutive health check failures
- Interval:** 30 seconds
- Success codes:** 200

30. Triage Target Group — Registered Targets

signaldock-triage-tg registered targets on port 8082.

The screenshot shows the AWS Management Console interface for Target groups. The left sidebar contains navigation links for various AWS services. The main content area displays a list of target groups under the 'signaldock-triage-tg' group. The table below shows the details of the registered targets.

Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
signaldock-api-tg	arn:aws:elasticloadbalancing:us-east-1:922419543079:targetgroup/signaldock-api-tg/1b319433bde91f69	8081	HTTP	Instance	signaldock-alb	vpc-02a85141e4fc122c8
signaldock-triage-tg	arn:aws:elasticloadbalancing:us-east-1:922419543079:targetgroup/signaldock-triage-tg/1b319433bde91f69	8082	HTTP	Instance	signaldock-alb	vpc-02a85141e4fc122c8
signaldock-web-tg	arn:aws:elasticloadbalancing:us-east-1:922419543079:targetgroup/signaldock-web-tg/1b319433bde91f69	8080	HTTP	Instance	signaldock-alb	vpc-02a85141e4fc122c8

Below the table, the 'Health check settings' for the 'signaldock-triage-tg' target group are displayed:

- Protocol:** HTTP
- Path:** /api/health
- Port:** Traffic port
- Healthy threshold:** 5 consecutive health check successes
- Unhealthy threshold:** 2 consecutive health check failures
- Interval:** 30 seconds
- Success codes:** 200

31. Target Groups — Full List

Target groups list showing all three target groups: web, api, and triage.

The screenshot shows the AWS Management Console interface for the 'Target groups' page. The left sidebar contains navigation links for various AWS services. The main content area displays a list of target groups. The 'signaldock-triage-tg' target group is selected, and the 'Targets' tab is active, showing two registered targets.

Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
signaldock-api-tg	arn:aws:elasticloadbalancing:us-east-1:123456789012:targetgroup/signaldock-api-tg/12345678901234567890	8081	HTTP	Instance	signaldock-alb	vpc-02a85141e4fc122c8
signaldock-triage-tg	arn:aws:elasticloadbalancing:us-east-1:123456789012:targetgroup/signaldock-triage-tg/12345678901234567890	8082	HTTP	Instance	signaldock-alb	vpc-02a85141e4fc122c8
signaldock-web-tg	arn:aws:elasticloadbalancing:us-east-1:123456789012:targetgroup/signaldock-web-tg/12345678901234567890	8080	HTTP	Instance	signaldock-alb	vpc-02a85141e4fc122c8

Target group: signaldock-triage-tg

Registered targets (2)

Instance ID	Name	Port	Zone	Health status	Health status details	Administrative override	Override details	Launch time	Anomaly detection
i-0c32f577816796031	-	8082	us-east-1b (us-east-1)	Healthy	-	No override	No override is currently active...	September 17, 2026, 19:09 (UTC-4)	Normal
i-0edaf118b7acdf764b	-	8082	us-east-1a (us-east-1)	Healthy	-	No override	No override is currently active...	September 17, 2026, 19:09 (UTC-4)	Normal

32. Triage Target Group — Health Check Settings

signaldock-triage-tg health check configuration, using the /triage/health path.

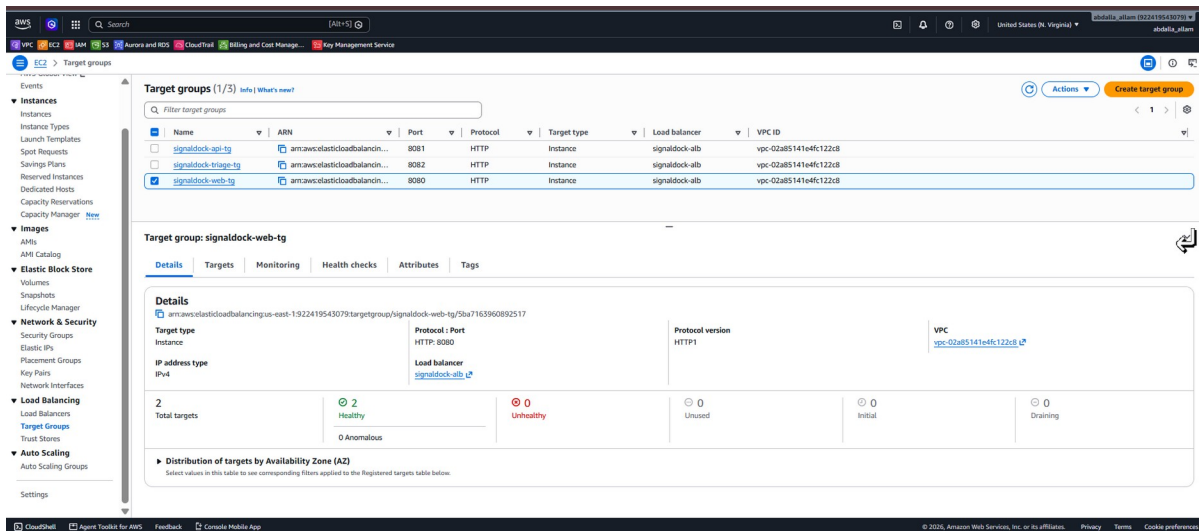
The screenshot shows the AWS Management Console interface for the 'Target groups' page. The 'signaldock-triage-tg' target group is selected, and the 'Health checks' tab is active, showing the health check settings.

Health check settings

Protocol	Path	Port	Healthy threshold
HTTP	/triage/health	Traffic port	5 consecutive health check successes
Unhealthy threshold	Timeout	Interval	Success codes
2 consecutive health check failures	5 seconds	30 seconds	200

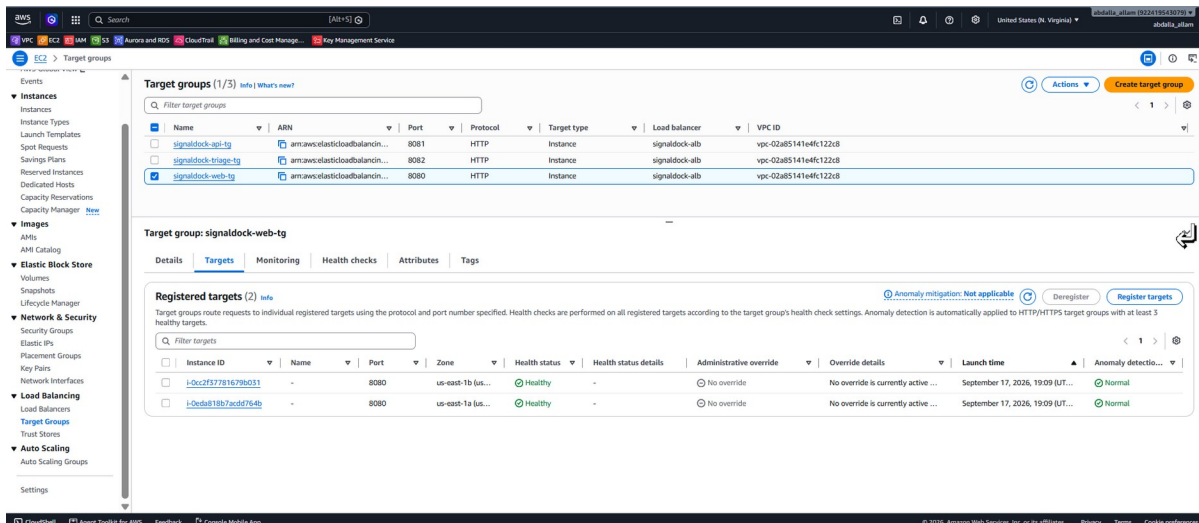
33. Web Target Group — Registered Targets

signaldock-web-tg registered targets on port 8080.



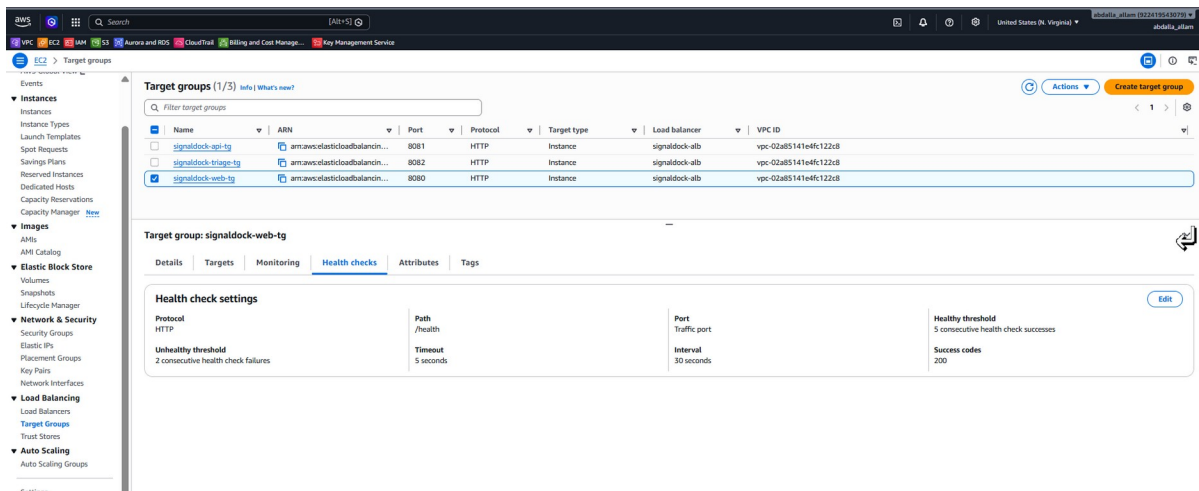
34. Web Target Group — Targets Detail

signaldock-web-tg registered targets, confirming both application nodes are healthy.



35. Web Target Group — Health Check Settings

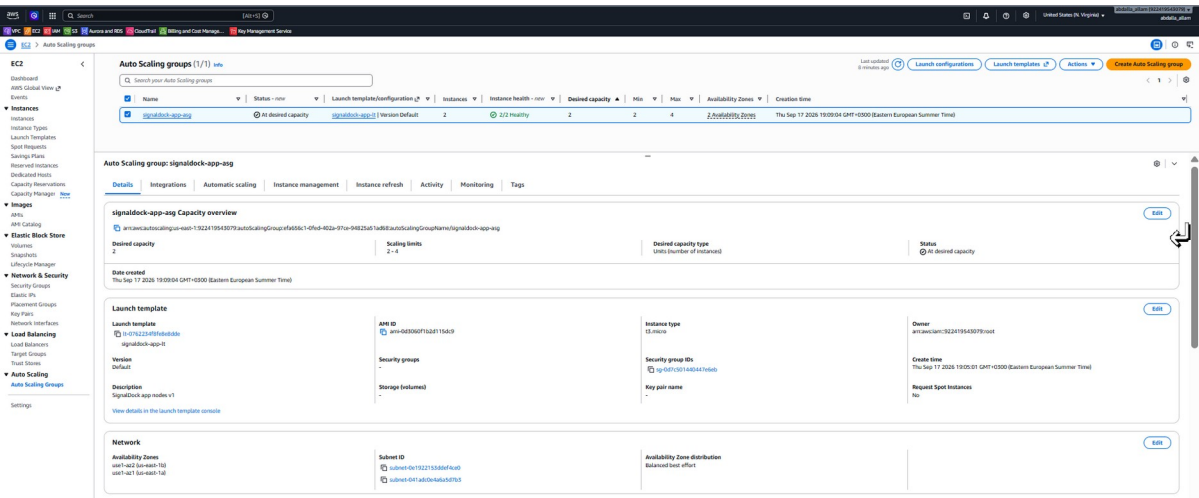
signaldock-web-tg health check configuration, using the /health path.



Section 8 — Auto Scaling Group

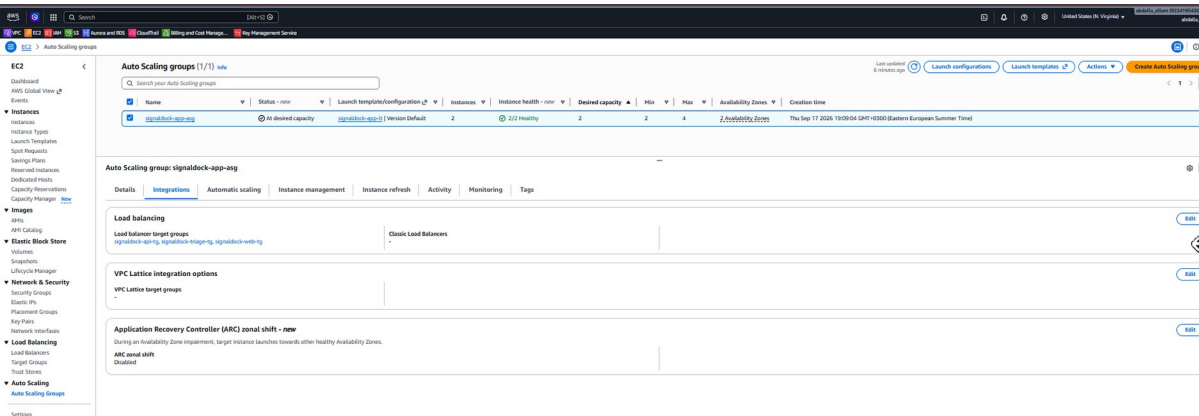
36. Auto Scaling Group — Overview

signaldock-app-asp overview, showing capacity and instance details.



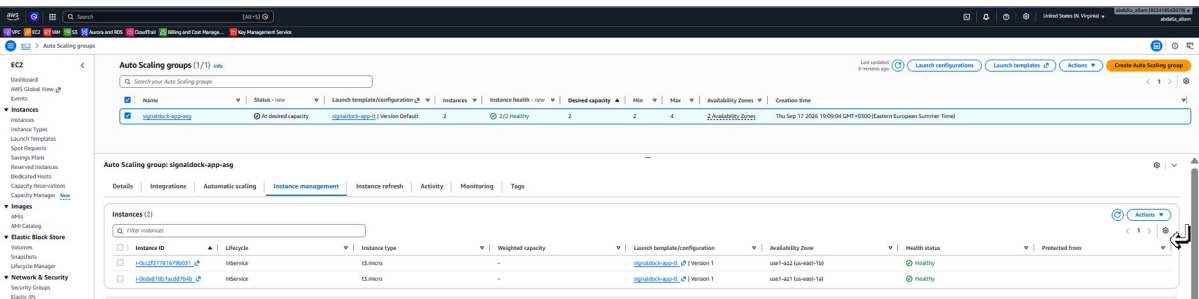
37. Auto Scaling Group — Configuration Detail

signaldock-app-asp configuration detail, including health check and scaling settings.



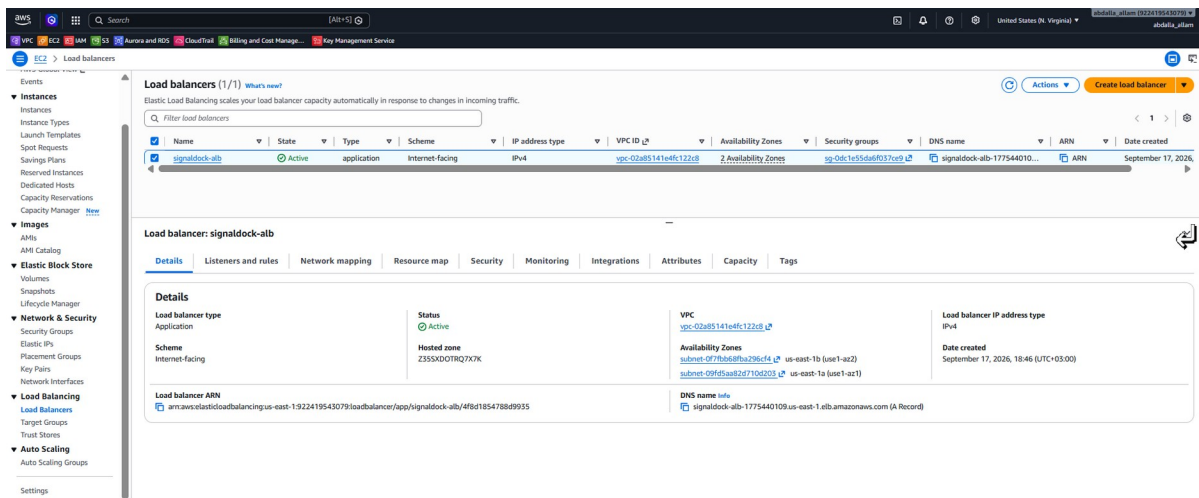
38. Auto Scaling Groups — List & Activity

Auto Scaling Groups list with the activity history for signaldock-app-asp.



39. Auto Scaling Group — Activity Tab

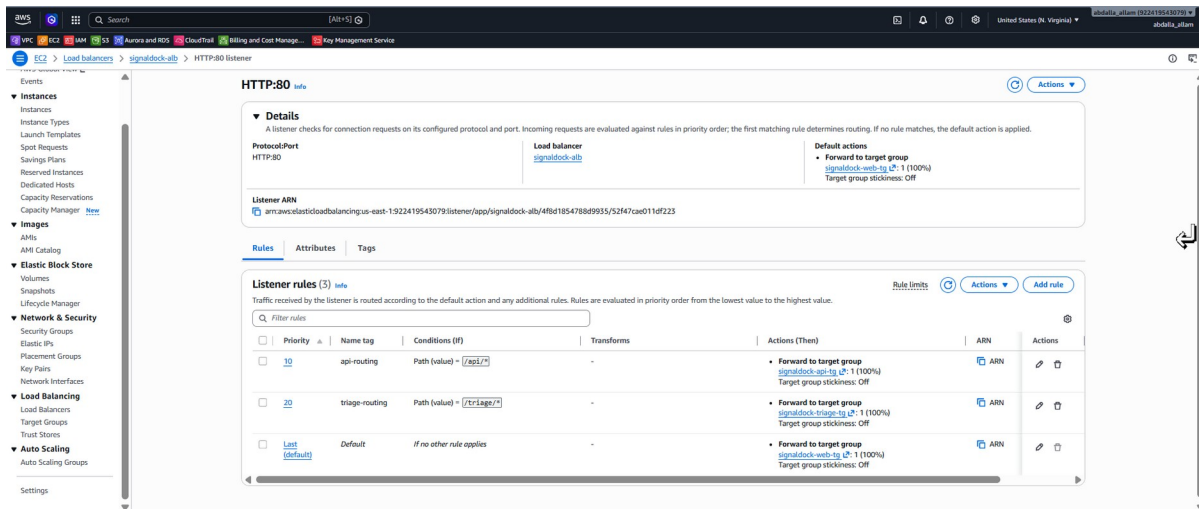
signaldock-app-asp Activity tab, showing scaling and health-check events.



Section 9 — Application Load Balancer & Path-Based Routing

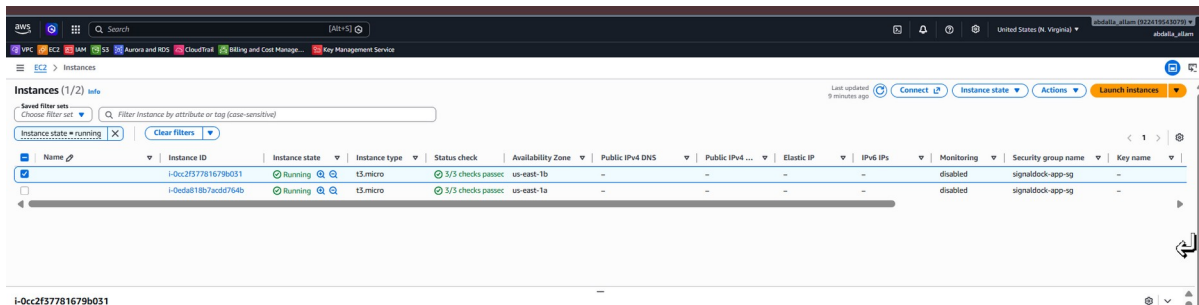
40. Load Balancers — signaldock-alb

Load Balancers list showing signaldock-alb.



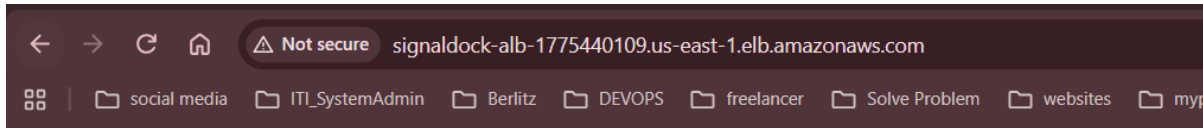
41. ALB — Path-Based Listener Rules

signaldock-alb listener rules, forwarding /, /api/*, and /trriage/* to their respective target groups.



42. Browser — SignalDock Web Service

SignalDock's Indicator Intake Portal served through the ALB DNS name, confirming the default path routes to the Web Service.



SignalDock

Indicator Intake Portal

Indicator

Type

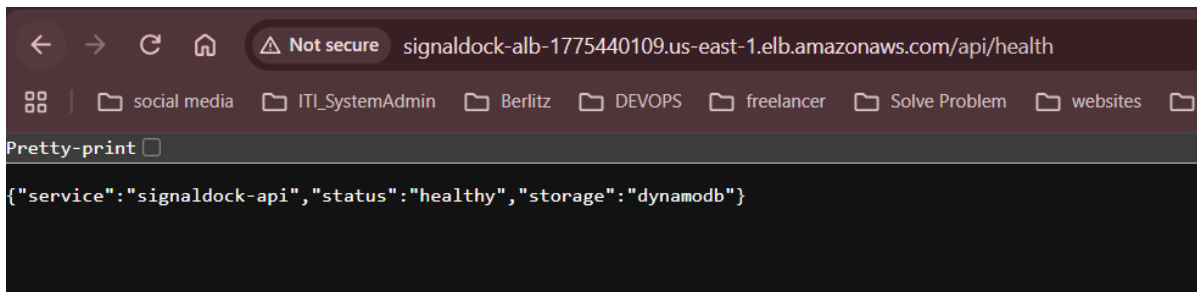
Severity

Created: sig-1789662976059

Section 10 — End-to-End Application Validation

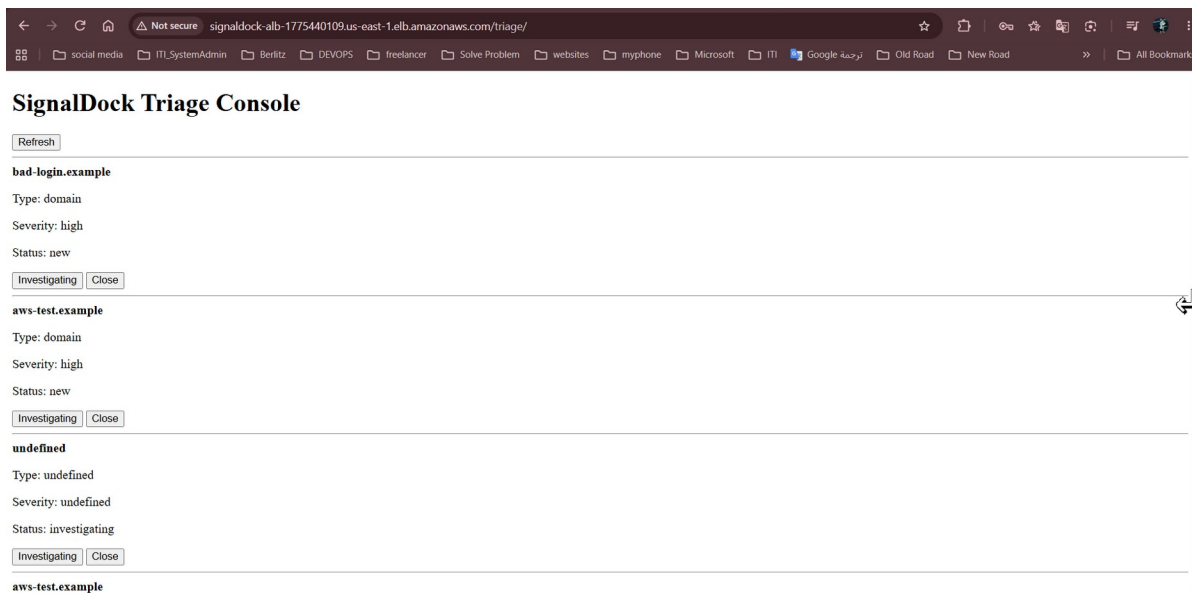
43. Browser — API Health via ALB

/api/health served through the ALB, confirming the storage mode is dynamodb in the AWS environment.



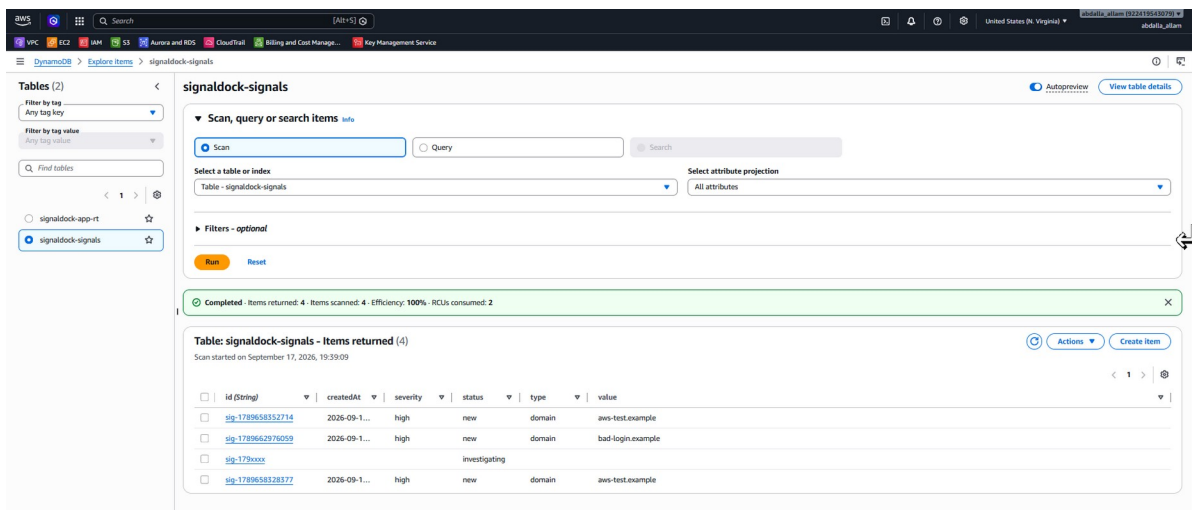
44. Browser — Triage Console

SignalDock's Triage Console, listing submitted indicators for analyst review.



45. DynamoDB — Stored Indicators

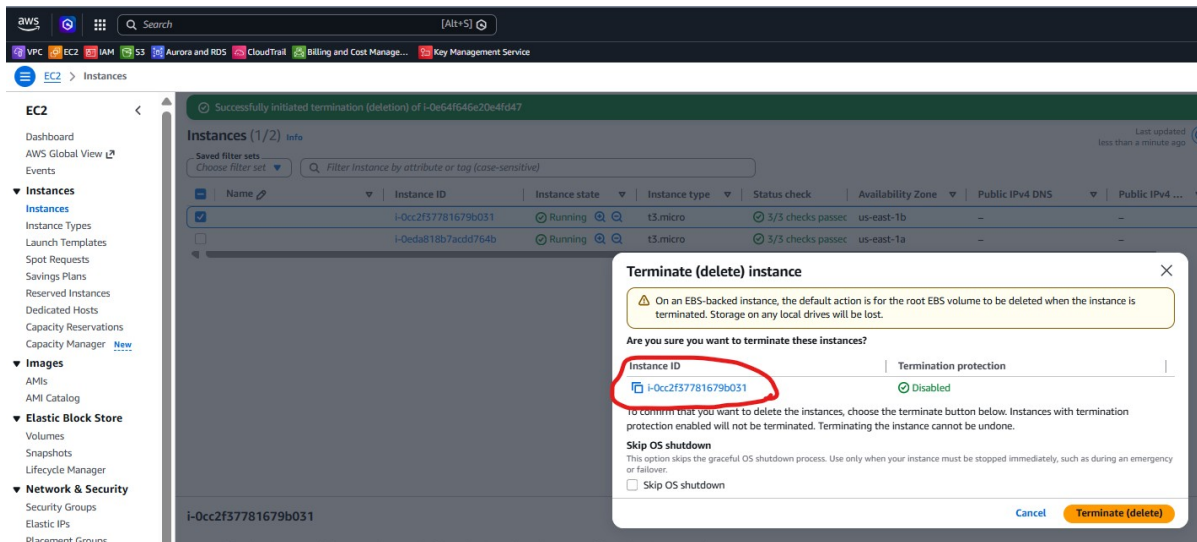
DynamoDB console scan of signaldock-signals, confirming submitted indicators are persisted.



Section 11 — Failure Testing: EC2 Replacement & Private Management

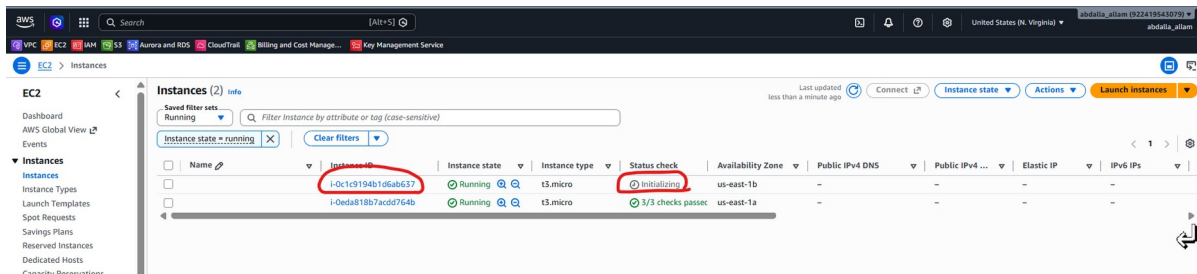
46. EC2 — Terminate Instance

Terminate Instance confirmation dialog, simulating an application node failure.



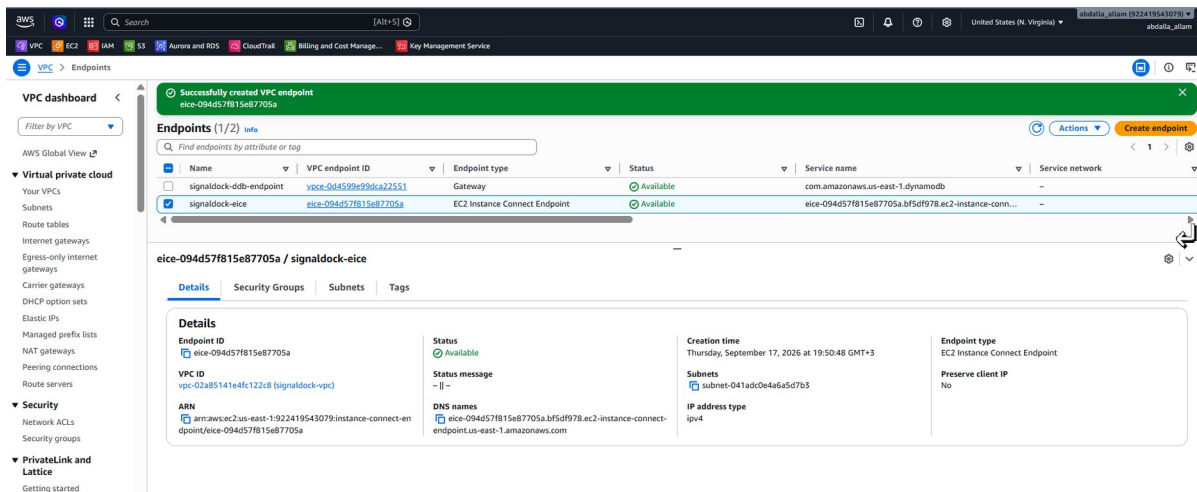
47. EC2 Instances — Replacement Launching

Instances list showing the terminated node shutting down while Auto Scaling launches a replacement.



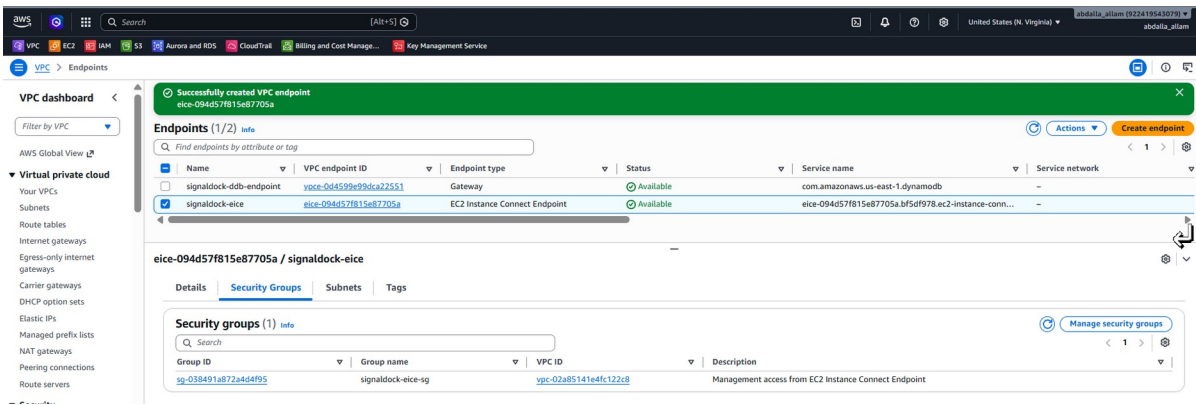
48. EC2 Instance Connect Endpoint — Created

Confirmation banner after creating the signaldock-eice EC2 Instance Connect Endpoint.



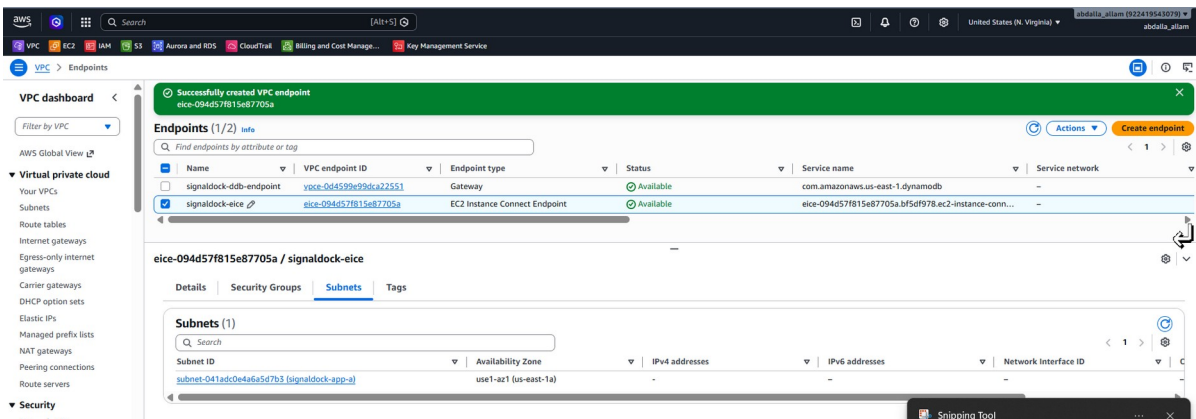
49. EC2 Instance Connect Endpoint — Security Groups Tab

signaldock-eice endpoint detail, Security Groups tab.



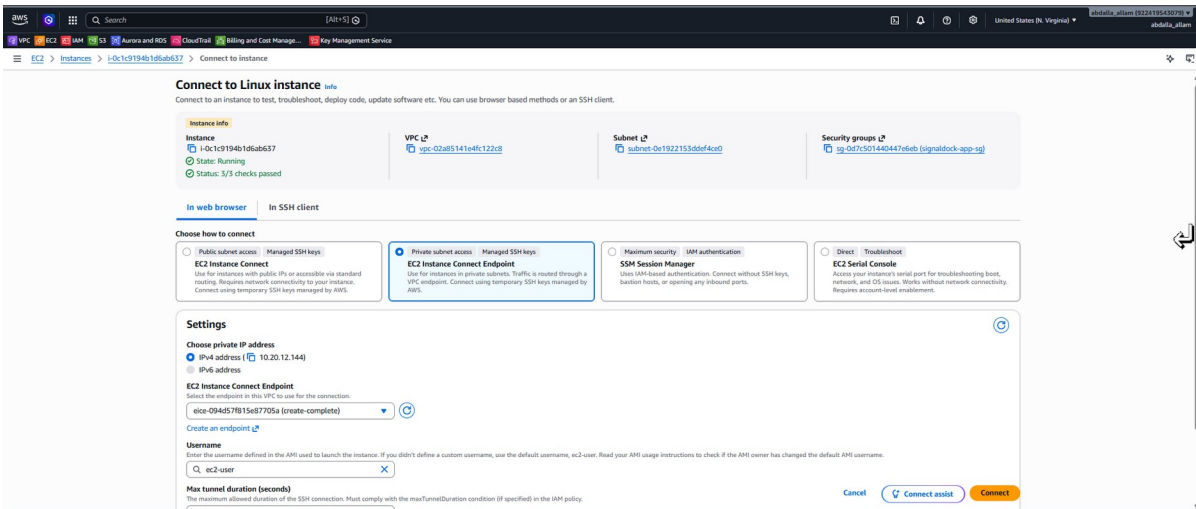
50. Connect to Linux Instance — via EICE

Connect to Linux Instance dialog, using the EC2 Instance Connect Endpoint for private access.



51. Terminal — Containers on the Replacement Instance

Confirming docker ps on the replacement instance shows all three containers healthy after Auto Scaling recovery.



52. Terminal — Recovery Validation

Validating connectivity and output on the replacement instance.

```
aws
[ec2-user@ip-10-20-12-144 ~]$ hostname
ip-10-20-12-144.ec2.internal
[ec2-user@ip-10-20-12-144 ~]$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 12:ef:e1:e3:46:eb brd ff:ff:ff:ff:ff:ff
    altname emp0s5
    altname eni-020e0a443071efeb0
    altname device-number-0.0
    inet 10.20.12.144/24 metric 512 brd 10.20.12.255 scope global dynamic ens5
        valid_lft 2484sec preferred_lft 2484sec
    inet6 fe80::10ff:fe1f:fe01:46eb/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
3: docker0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether 02:42:2d:68:4d:33 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
    inet6 fe80::42:2d:68:4d:33/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
5: veth8e33f5d84f4: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker0 state UP group default
    link/ether 96:76:b4:0b:74:72 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet6 fe80::9676b40b7472/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
7: veth8b96d94fe: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker0 state UP group default
    link/ether 9a:c4:4f:7b:a6:c3 brd ff:ff:ff:ff:ff:ff link-netnsid 1
    inet6 fe80::9ac44f7ba6c3/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
9: veth8e96b84f0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker0 state UP group default
    link/ether 62:1b:7e:bd:60:67 brd ff:ff:ff:ff:ff:ff link-netnsid 2
    inet6 fe80::621b7ebd6067/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
[ec2-user@ip-10-20-12-144 ~]$ docker ps
CONTAINER ID        IMAGE               COMMAND                  CREATED      STATUS      PORTS
2464abbb4210      signaldock-triage:1.0      "docker-entrypoint.s..." 2 hours ago  Up 18 minutes      0.0.0.0:8082->8082/tcp, :::8082->8082/tcp      signaldock-triage
22f9f8be0e01      signaldock-api:1.0        "docker-entrypoint.s..." 2 hours ago  Up 18 minutes      0.0.0.0:8081->8081/tcp, :::8081->8081/tcp      signaldock-api
ea957dfb4abb      signaldock-web:1.0        "docker-entrypoint.s..." 2 hours ago  Up 18 minutes      0.0.0.0:8080->8080/tcp, :::8080->8080/tcp      signaldock-web
[ec2-user@ip-10-20-12-144 ~]$ docker ps --format "table {{.Names}}\t{{.Ports}}"
NAME                PORTS
signaldock-triage    0.0.0.0:8082->8082/tcp, :::8082->8082/tcp
signaldock-api       0.0.0.0:8081->8081/tcp, :::8081->8081/tcp
signaldock-web       0.0.0.0:8080->8080/tcp, :::8080->8080/tcp
[ec2-user@ip-10-20-12-144 ~]$ curl http://localhost:8080/health
ok[ec2-user@ip-10-20-12-144 ~]$ curl http://localhost:8081/api/health
{"service":"signaldock-api","status":"healthy","storage":"dynamodb"}[ec2-user@ip-10-20-12-144 ~]$ curl http://localhost:8082/triage/health
ok[ec2-user@ip-10-20-12-144 ~]$ curl http://localhost:8081/api/signal-1
{"value":"bad-login.example","createdAt":"2026-09-17T16:36:16.059Z","id":"sig-1789662976093","type":"domain","severity":"high","status":"new"},{"value":"aws-test.example","id":"sig-1789658282714","severity":"high","status":"new"},{"value":"aws-test.example","id":"sig-1789658282714","severity":"high","status":"new","type":"domain"},{"id":"sig-1789xxx","status":"investigating"},{"createdAt":"2026-09-17T15:18:48.377Z","value":"aws-test.example","id":"sig-178965828282777","status":"new","severity":"high","type":"domain"}[ec2-user@ip-10-20-12-144 ~]$
```

53. Browser — Submitting After Recovery

Submitting a new indicator after recovery to confirm the application is fully functional on the replacement node.

aws

Search

[Alt+S]

United States (N. Virginia)

abdalla_allam (922419543079)

abdalla_allam

VPC

EC2

IAM

S3

Aurora and RDS

CloudTrail

Billing and Cost Manage...

Key Management Service

[ec2-user@ip-10-20-12-144 ~]\$ docker ps

CONTAINER ID	IMAGE	NAMES	COMMAND	CREATED	STATUS	PORTS
2464a5bb4210	signaldock-triage:1.0	signaldock-triage	"docker-entrypoint.s..."	2 hours ago	Up 27 minutes	0.0.0.0:8082->8082/tcp, :::8082->8082/tcp
22f9ffbe0e01	signaldock-api:1.0	signaldock-api	"docker-entrypoint.s..."	2 hours ago	Up 27 minutes	0.0.0.0:8081->8081/tcp, :::8081->8081/tcp
ea957dfb4abb	signaldock-web:1.0	signaldock-web	"docker-entrypoint.s..."	2 hours ago	Up 27 minutes	0.0.0.0:8080->8080/tcp, :::8080->8080/tcp

[ec2-user@ip-10-20-12-144 ~]\$ docker stop signaldock-web

signaldock-web

[ec2-user@ip-10-20-12-144 ~]\$

[ec2-user@ip-10-20-12-144 ~]\$ docker ps

CONTAINER ID	IMAGE	NAMES	COMMAND	CREATED	STATUS	PORTS
2464a5bb4210	signaldock-triage:1.0	signaldock-triage	"docker-entrypoint.s..."	2 hours ago	Up 28 minutes	0.0.0.0:8082->8082/tcp, :::8082->8082/tcp
22f9ffbe0e01	signaldock-api:1.0	signaldock-api	"docker-entrypoint.s..."	2 hours ago	Up 28 minutes	0.0.0.0:8081->8081/tcp, :::8081->8081/tcp

[ec2-user@ip-10-20-12-144 ~]\$ curl --connect-timeout 3 http://localhost:8080/health

curl: (7) Failed to connect to localhost:8080 after 0 ms: Could not connect to server

[ec2-user@ip-10-20-12-144 ~]\$

i-0c1c9194b1d6ab637

PrivateIPs: 10.20.12.144

54. Web Target Group — Targets After Recovery

signaldock-web-tg registered targets, confirming the replacement instance is healthy.

aws

Search

[Alt+S]

United States (N. Virginia)

abdalla_allam (922419543079)

abdalla_allam

VPC

EC2

IAM

S3

Aurora and RDS

CloudTrail

Billing and Cost Manage...

Key Management Service

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Load Balancers

Target Groups

Trust Stores

Auto Scaling

Auto Scaling Groups

Target groups

Target groups (1/3)

Filter target groups

Name	ARN	Port	Protocol	Target type	Load balancer	VPC ID
signaldock-api-tg	arn:aws:elasticloadbalancing:us-east-1b:922419543079:targetgroup/signaldock-api-tg/2464a5bb4210	8081	HTTP	Instance	signaldock-alb	vpc-02a85141e4fc122c8
signaldock-triage-tg	arn:aws:elasticloadbalancing:us-east-1b:922419543079:targetgroup/signaldock-triage-tg/22f9ffbe0e01	8082	HTTP	Instance	signaldock-alb	vpc-02a85141e4fc122c8
signaldock-web-tg	arn:aws:elasticloadbalancing:us-east-1b:922419543079:targetgroup/signaldock-web-tg/ea957dfb4abb	8080	HTTP	Instance	signaldock-alb	vpc-02a85141e4fc122c8

Target group: signaldock-web-tg

DetailsTargetsMonitoringHealth checksAttributesTags

Registered targets (2)

Anomaly mitigation: Not applicable

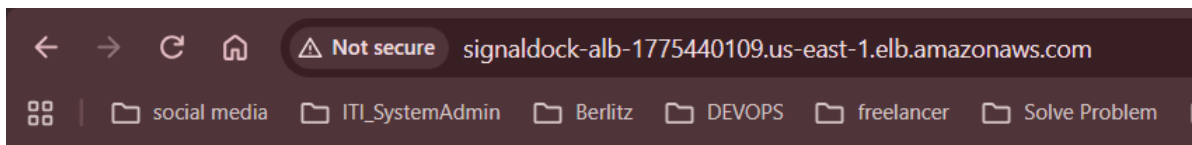
DeregisterRegister targets

Filter targets

Instance ID	Name	Port	Zone	Health status	Health status details	Admini...	Overri...	Launch...	Anomaly detection result
i-0c1c9194b1d6ab637	-	8080	us-east-1b (us-east-1)	Unhealthy	Health checks failed	No override	No overri...	Septembe...	Normal
i-0eda818b7acd0764b	-	8080	us-east-1a (us-east-1)	Healthy	-	No override	No overri...	Septembe...	Normal

55. EC2 Instances — Back to Baseline

Instances list confirming the environment returned to two healthy application nodes.



SignalDock

Indicator Intake Portal

Indicator

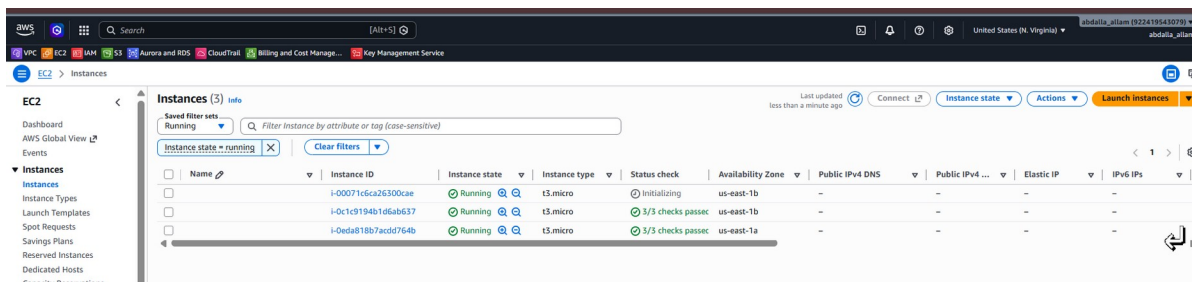
Type

Severity

Section 12 — Dynamic Auto Scaling (CPU Target Tracking)

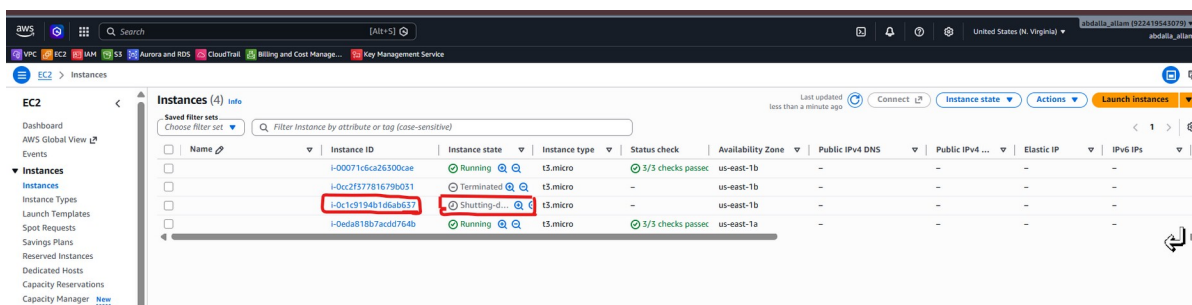
56. EC2 Instances — Before Scaling Test

Instances list before the dynamic scaling test.



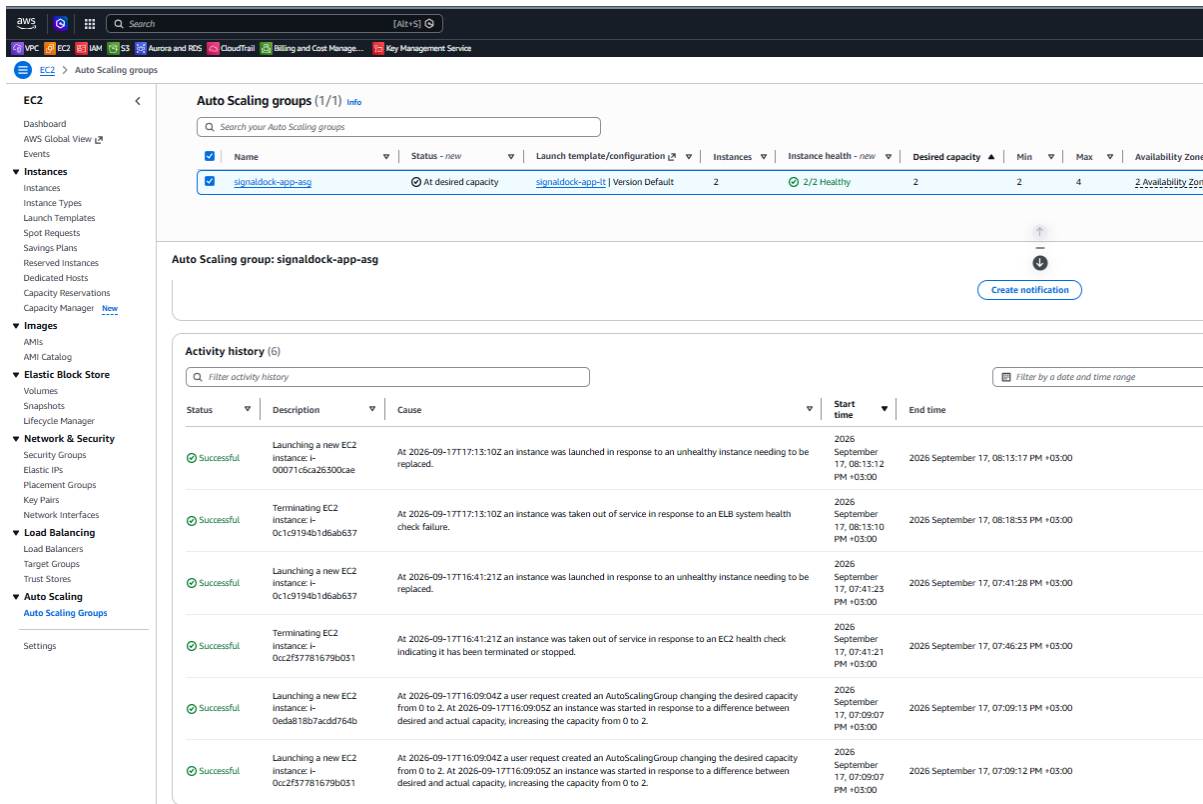
57. Auto Scaling Group — Activity Tab

signaldock-app-asg Activity tab, reviewed ahead of the scaling test.



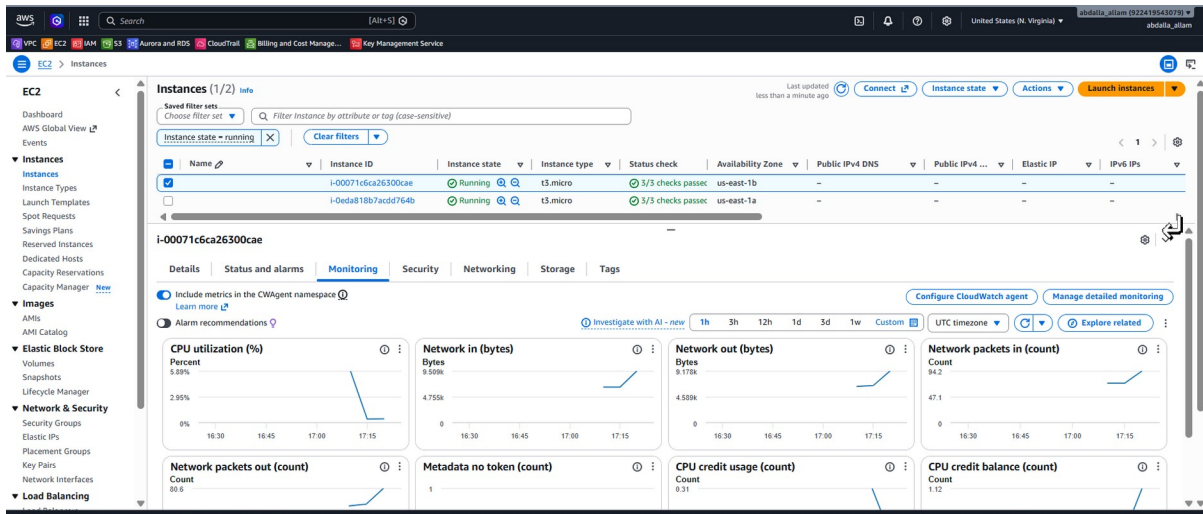
58. EC2 Instance — Monitoring Tab

Per-instance Monitoring tab showing CPU utilization and network graphs.



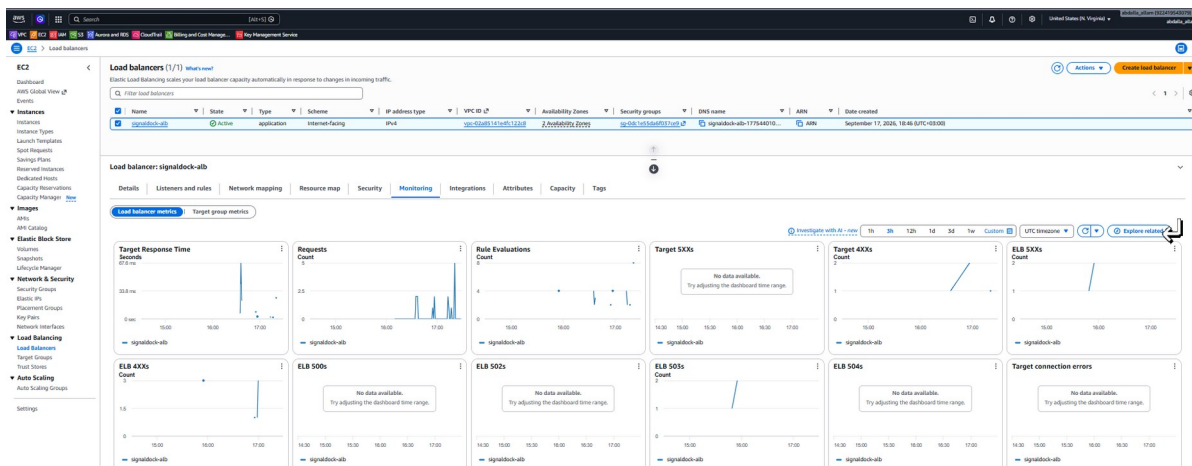
59. Auto Scaling Group — Monitoring Tab

signaldock-app-asg Monitoring tab, graphing group-level metrics.



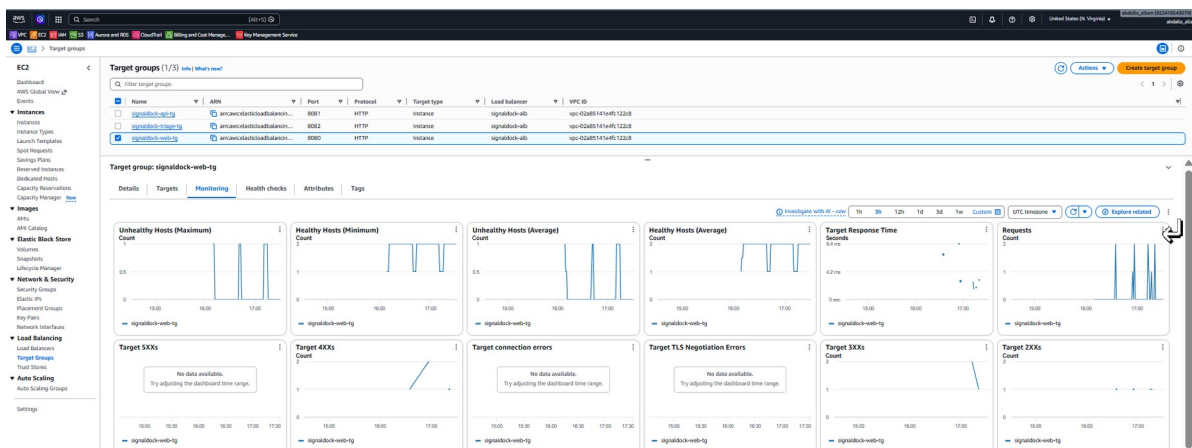
60. Target Group — Monitoring Tab

Target group Monitoring tab, graphing request and health metrics.



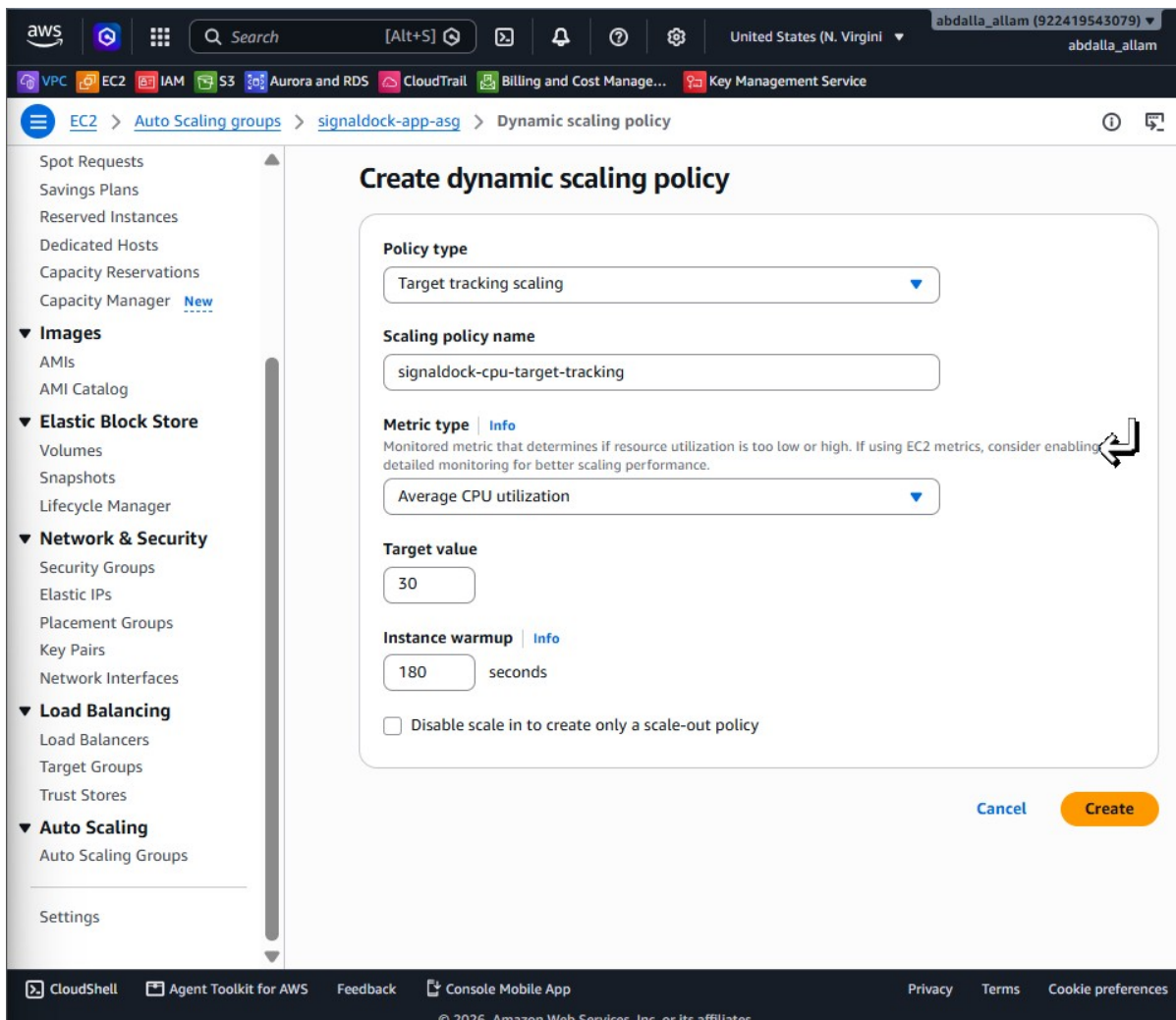
61. Creating the Dynamic Scaling Policy

Creating a target tracking scaling policy based on Average CPU Utilization.



62. Terminal — Baseline CPU (top)

Baseline CPU utilization on an application node before the load test, viewed with top.



63. Terminal — Elevated CPU (top)

Elevated CPU utilization during the stress test, viewed with top.

aws

Search

[Alt+S]

United States (N. Virginia)

abdalla_allam (922419543079)

abdalla_allam

VPC

EC2

IAM

S3

Aurora and RDS

CloudTrail

Billing and Cost Manage...

Key Management Service

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
4216	root	20	0	224060	3572	2860	R	6.2	0.4	0:00.01	top
1	root	20	0	173572	17832	10928	S	0.0	1.9	0:01.37	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-xfs-log/nvme0n1p1
12	root	20	0	0	0	0	I	0.0	0.0	0:00.29	kworker/u8:0-events_unbound
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	S	0.0	0.0	0:00.03	ksoftirqd/0
15	root	20	0	0	0	0	I	0.0	0.0	0:00.06	rcu_preempt
16	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_par_gp_kthread_worker/0
17	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_gp_kthread_worker
18	root	rt	0	0	0	0	S	0.0	0.0	0:00.01	migration/0
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
20	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
21	root	rt	0	0	0	0	S	0.0	0.0	0:00.04	migration/1
22	root	20	0	0	0	0	S	0.0	0.0	0:00.02	ksoftirqd/1
24	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/1:0H-events_highpri
27	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kdevtmpfs
28	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-inet_frag_wq
29	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
30	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthread
31	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_trace_kthread
32	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kauditd
34	root	20	0	0	0	0	S	0.0	0.0	0:00.00	khungtaskd
35	root	20	0	0	0	0	S	0.0	0.0	0:00.00	oom_reaper
37	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-writeback
38	root	20	0	0	0	0	S	0.0	0.0	0:00.08	kcompactd0

```
[root@ip-10-20-12-148 ~]# yes > /dev/null &
[1] 4217
[root@ip-10-20-12-148 ~]# yes > /dev/null &
[2] 4218
[root@ip-10-20-12-148 ~]# top
```

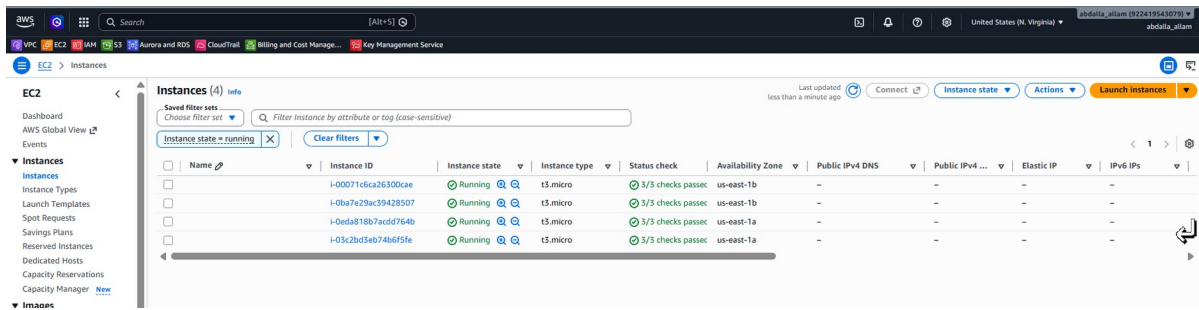
i-00071c6ca26300cae

PrivatelPs: 10.20.12.148

X

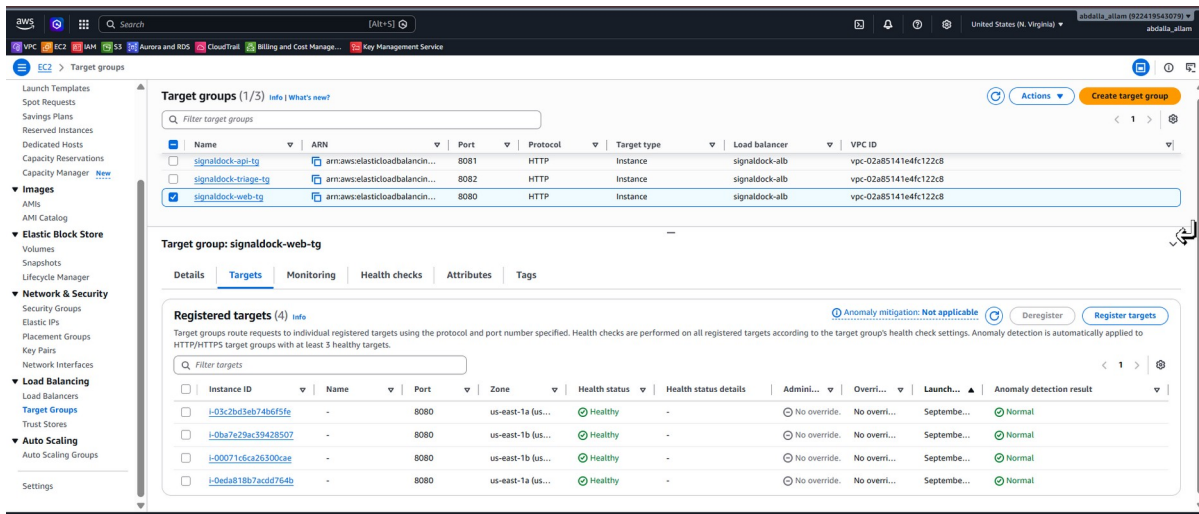
64. EC2 Instances — Scaling Out

Instances list showing an additional application node launching during scale-out.



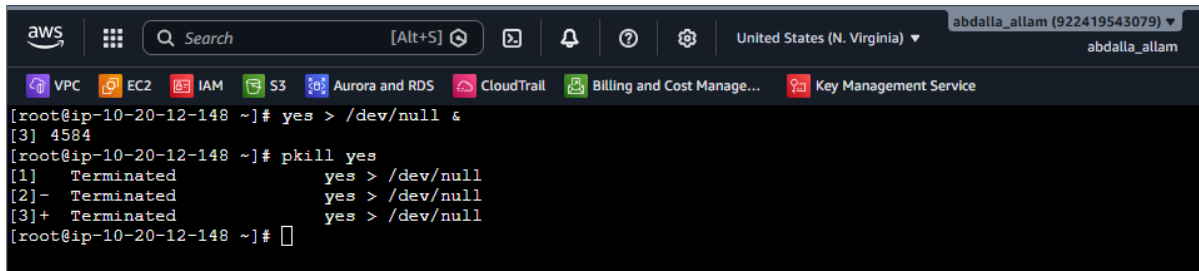
67. Terminal — Stopping the CPU Load

Stopping the CPU load test with pkill yes.



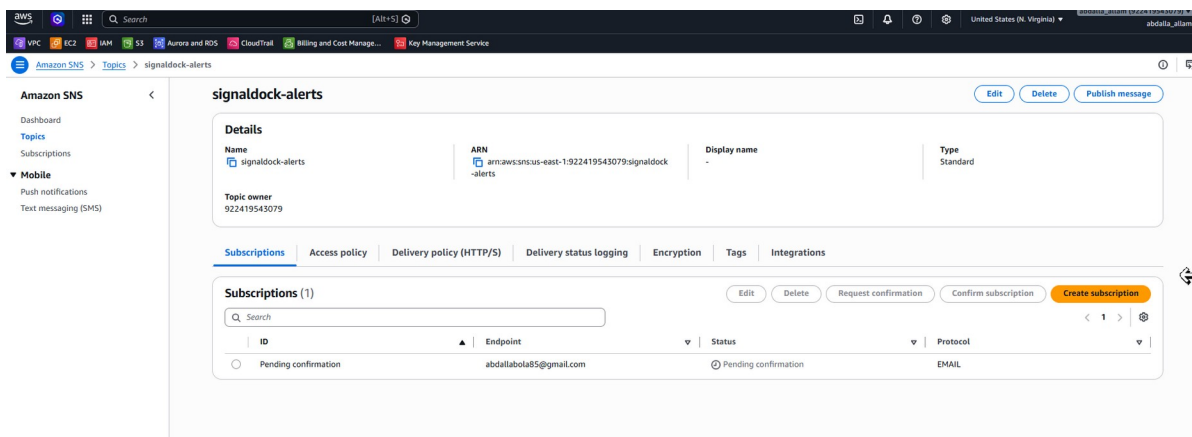
68. Terminal — CPU Back to Normal (top)

CPU utilization back to normal after the load test, viewed with top.



69. EC2 Instances — Scaling In

Instances list showing the group scaling back in toward baseline capacity.



74. SNS Subscription Confirmed

Email subscription to signaldock-alerts, confirmed and ready to receive alarm notifications.



Simple Notification Service

Subscription confirmed!

You have successfully subscribed.

Your subscription's id is:

`arn:aws:sns:us-east-1:922419543079:signaldock-alerts:7dfa125c-7262-48b7-b024-8d113ea0ad6f`

If it was not your intention to subscribe, [click here to unsubscribe](#).

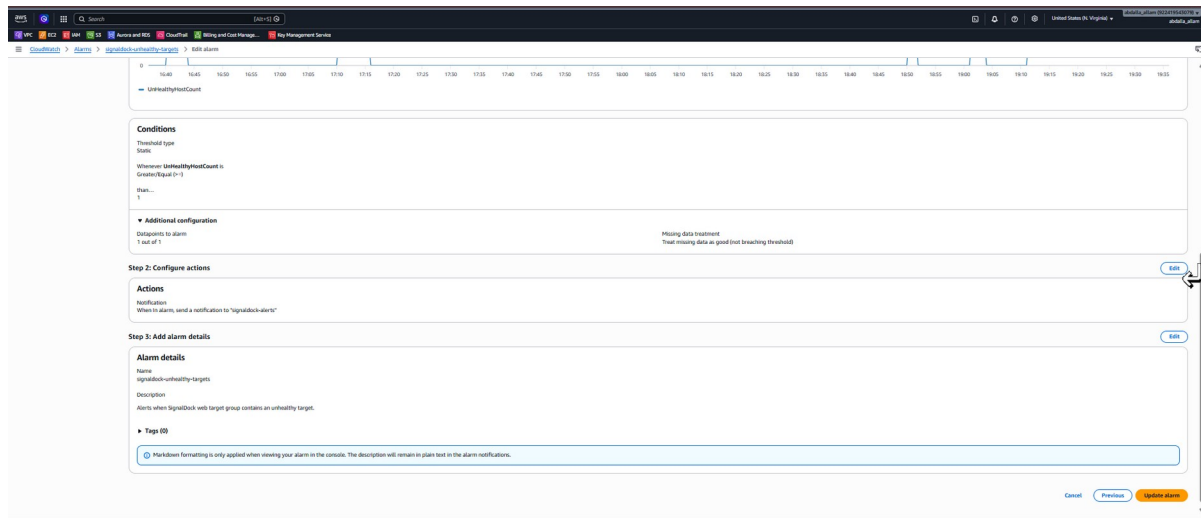
75. CloudWatch Alarm — Metric Preview

Creating the CloudWatch alarm: previewing the UnHealthyHostCount metric against the alarm threshold.



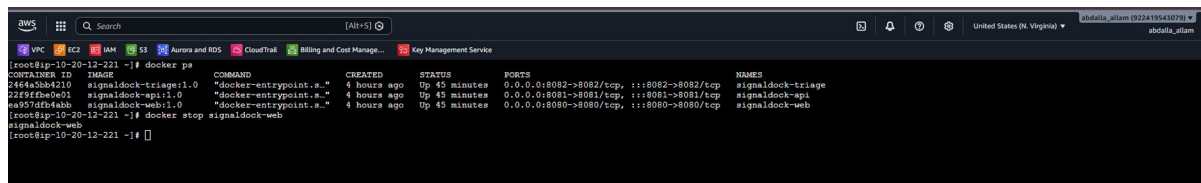
76. CloudWatch Alarm — Configure Actions

Configuring the alarm action to notify the signaldock-alerts SNS topic.



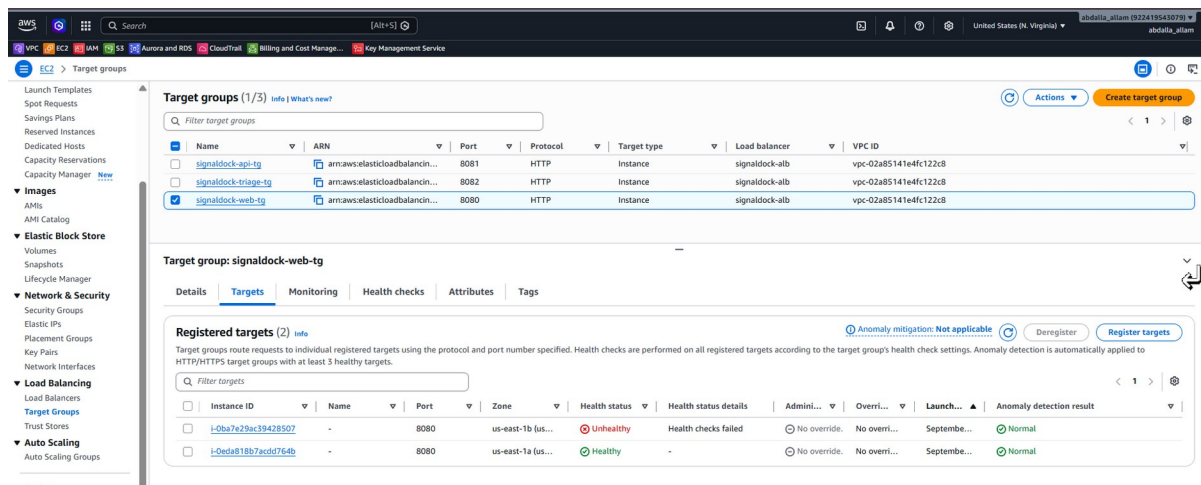
77. Terminal — Alarm Test Validation

Validating application/container state on the instance during the alarm test.



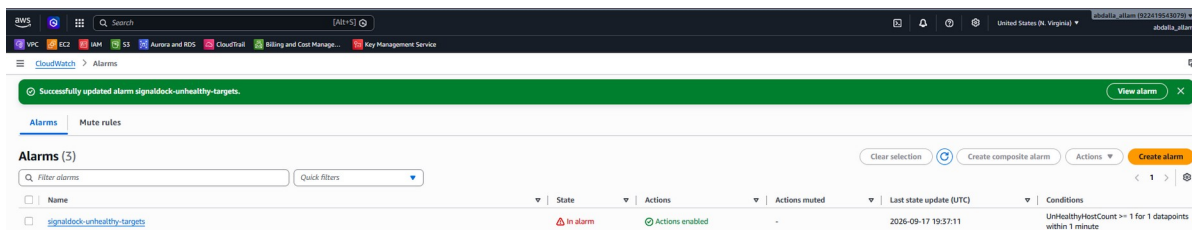
78. Web Target Group — Healthy After Alarm Test

signaldock-web-tg registered targets healthy again after the alarm test completed.



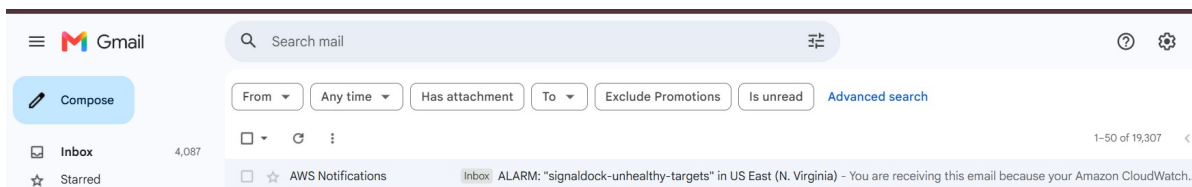
79. CloudWatch Alarms — Updated

CloudWatch Alarms list confirming signaldock-unhealthy-targets was successfully created/updated.



80. SNS Email — Alarm Notification

Email alert received via SNS: 'ALARM: signaldock-unhealthy-targets' in US East (N. Virginia).



Section 14 — CI: GitHub Actions Docker Build Validation

81. Creating the GitHub Actions Workflow

PowerShell terminal creating the .github/workflows directory and the docker-build.yml workflow file.

```
Windows PowerShell
PS E:\myproject\signaldock> mkdir .github

Directory: E:\myproject\signaldock

Mode                LastWriteTime         Length Name
----                -
d-----          9/17/2026  10:56 PM                .github

PS E:\myproject\signaldock> mkdir .github\workflows

Directory: E:\myproject\signaldock\.github

Mode                LastWriteTime         Length Name
----                -
d-----          9/17/2026  10:57 PM                workflows

PS E:\myproject\signaldock> New-Item .github\workflows\docker-build.yml

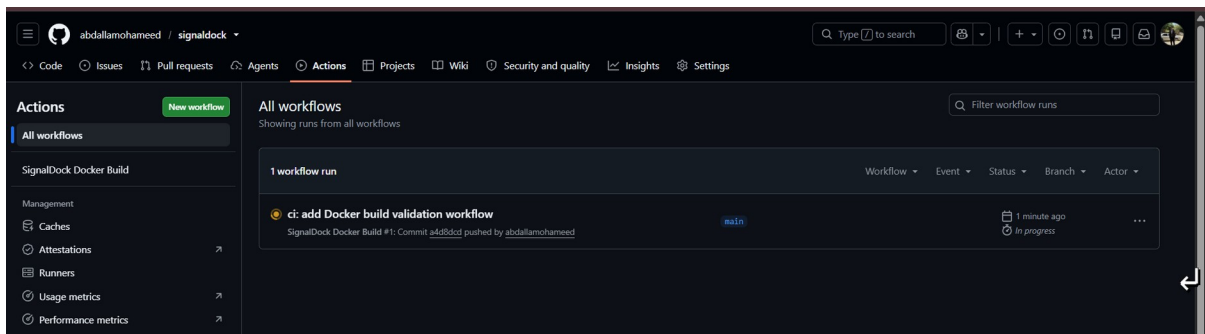
Directory: E:\myproject\signaldock\.github\workflows

Mode                LastWriteTime         Length Name
----                -
-a-----          9/17/2026  10:57 PM              0 docker-build.yml

PS E:\myproject\signaldock>
```

82. GitHub Actions — Workflow Listed

GitHub Actions showing the SignalDock Docker Build workflow after the CI commit.



83. GitHub Actions — Run Detail

Workflow run detail confirming all three Docker services (web, api, triage) build successfully in CI.

abdallamohamed / signaldock

Type [] to search

<> Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

<- SignalDock Docker Build

ci: add Docker build validation workflow #1

Re-run all jobs ...

Summary

All jobs

docker-build

Run details Usage Workflow file

> Annotations
1 warning

docker-build

succeeded 2 minutes ago in 17s

Search logs

Set up job 0s

Checkout repository 1s

Build Web service 7s

Build API service 3s

Build Triage service 3s

Show built images 0s

Post checkout repository 0s

Complete job 0s